



Data Warehouses

Lecture 6



Reminder

Last meeting:

- Data warehouse

- Data quality and profile

Lecture Goals

Goal:

Multidimensional model

Logical model – facts, measures and dimensions

What is a data cube?

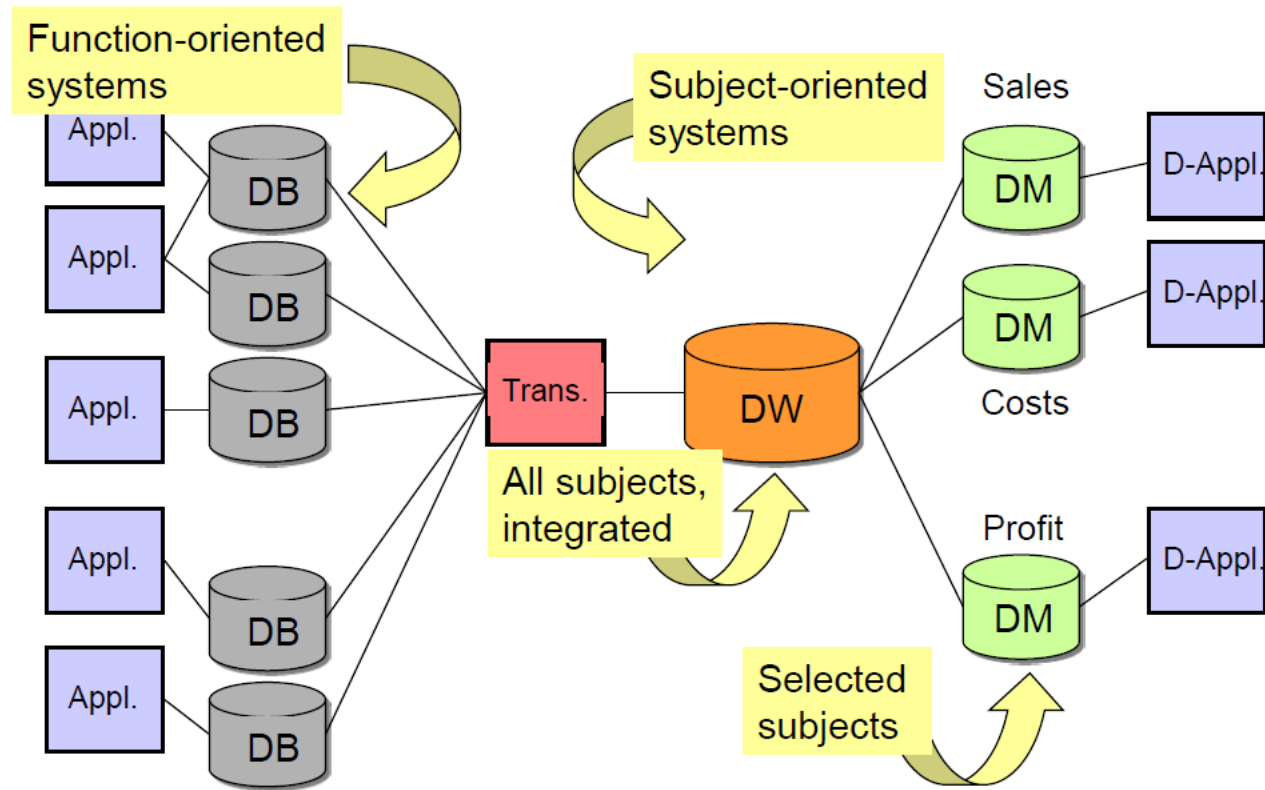
Drilling into data using dimensions

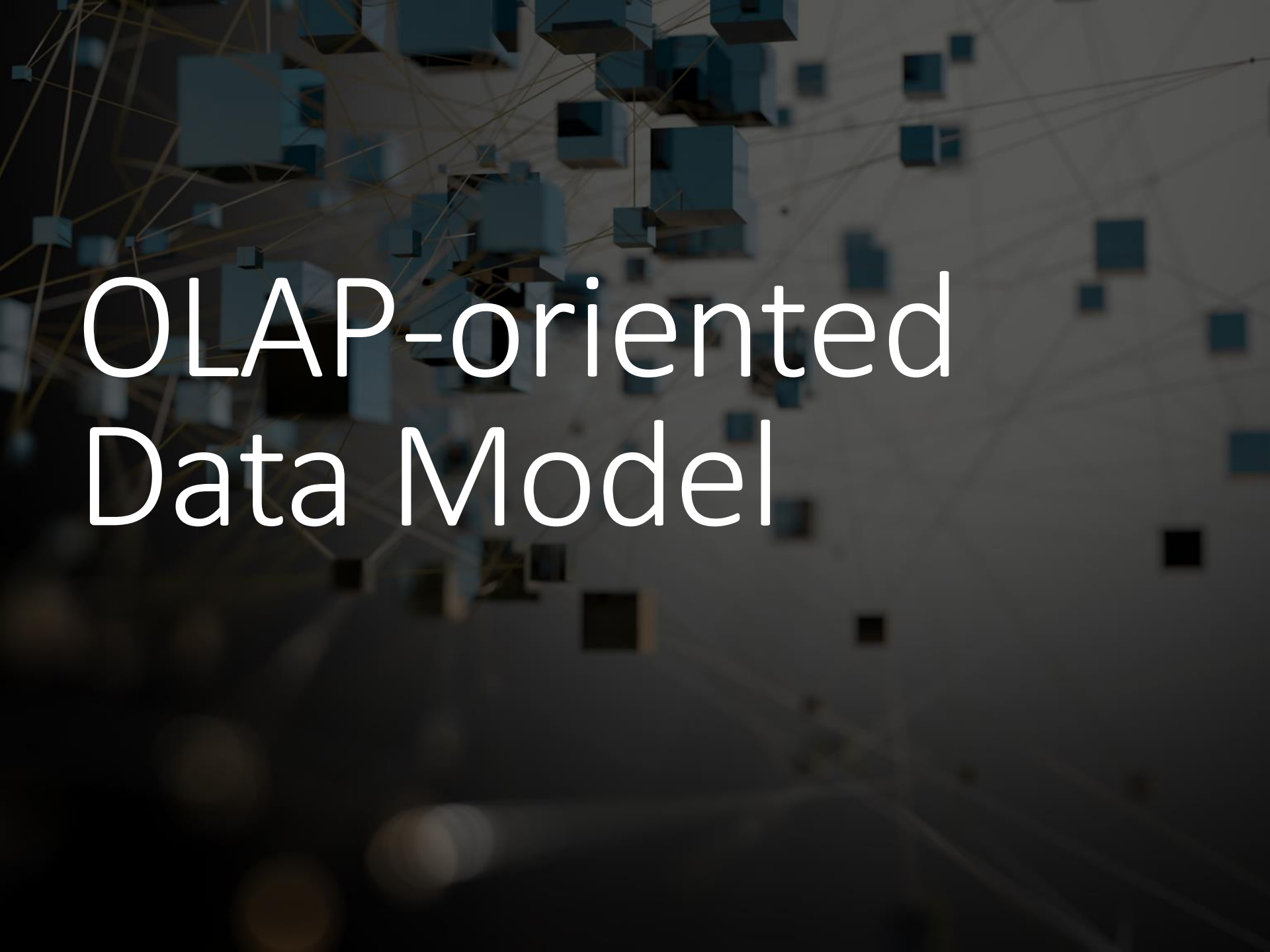
Measures and aggregates

OLAP operations

What can we do with a data cube?

Data warehouse



The background is a dark, textured surface with a network of thin, yellow lines connecting various blue, three-dimensional cubes. The cubes are scattered across the frame, some appearing in sharp focus while others are blurred, creating a sense of depth and complexity. The overall aesthetic is modern and technological.

OLAP-oriented Data Model

OLAP data

“Through measurement comes knowledge.”

Heike Kamerlingh Onnes



Business Advantage

Information delivery is possible,
when

Proper efficiency/quality measures are
defined

Summarize data

Support decisions

Proper questions are formulated

Did something happened?

Why something happened?

What is going to happen?

Unknown data dependencies/relations

Examples

Did something happened?

Is the decrease in sales during summer months this year comparable to average decrease in past years?

Why something happened?

Is the increase of sales amount of salesperson 202 due to general increase in sales?

What is going to happen?

Will the sales of our products increase before the upcoming holidays?

Unknown data dependencies/relations

Is there some correlation between customer's demographics and generate profit?

OLAP users

We sell products in various markets, and we measure our performance over time



Business Manager

OLAP users

Managers, Analysts, CEOs, etc.

Decision makers

Data is used for decision support

Data is consumed visually

OLAP needs numbers



The management of the organization is typically interested in aggregated data – specific numbers, like:

***Quantitative** comparison of product **sales values** over the years in the same region*

*What is the **total value of sales** of a given product in a specific store during the period*

*How much **money was spent** on a specific product promotion in a specific industry and in a specific city*



Multi-dimensional Model

Without some kind of context, a measurement is meaningless.

If you are told “*sales were \$10,000*”:

There is not much you can do with this information.

Is that sales of a single product, or many products?

Does it represent a single transaction, or the company’s total sales from conception to date?



Multi-dimensional Model

Dimensions provide context for facts.

To understand what “**\$10,000**” means, you need more information.

“Order dollars were \$10,000 for electronic products in January 2009.”

By adding dimensional context—a product category, a month, and a year—the fact has been made useful.

OLAP Needs

Consider the following business questions:

What are gross margins by product category for January?

What is the average account balance by education level?

What is the return rate by supplier?

Each focuses on a business process:

sales, account management, return processing

These process-centric questions do not focus on individual activities or transactions.

To answer them, it is necessary to look at a group of transactions.

OLAP Needs

Consider the following business questions:

What are gross margins by product category for January?

What is the average account balance by education level?

What is the return rate by supplier?

Most importantly, each of these questions reveals something about how its respective business process is measured.

The study of sales involves the measurement of gross margin.

Financial institutions measure account balance.

Purchasing managers watch the return quantities.

OLAP Needs

Consider the following business questions:

What are gross margins by product category for January?

What is the average account balance by education level?

What is the return rate by supplier?

Context, again, is revealed in business questions or reports.

In the exemplar questions:

Gross margin is viewed in the context of product categories and time (the month of January).

Sick days are viewed in the context of a department (marketing) and time (last year).

Payables are viewed in the context of their status (outstanding) and vendor.

OLAP Needs

Business processes

operational activities performed by the organization

such as taking an order, processing an insurance claim, enrolling students for class, or creating a snapshot of each account each month

managers are interested in capturing performance metrics

based on measured events

process selection

is important because it defines a specific design goal and allows you to declare grain, dimensions and facts.

Kimball Group

OLAP Data

“There are typically a number of **different dimensions** from which a given pool of data can be analysed.”

“This plural perspective, or **Multidimensional Conceptual View** appears to be the way most business persons naturally view their enterprise.”

E.F. Codd, S.B. Codd and C.T. Salley,
1993

Mental model of a dataset



Our mental model of a dataset changes the way we ask questions

the way how we ask questions results from a particular mental model we use

it's easier to comprehend data – if data model is „compatible“ with our mental model



Mental model of a dataset

whether we think of the data as a

- collection of rows of numbers that we can aggregate bottom-up,
- complete dataset that we can slice top-down to ask questions.
- etc.

Mental model of a dataset

At the beginning there were rows of data

Data at row-level

The numbers are there, listed, one by one

Suitable for low-level, detailed, focusing on individual data point workloads

In a way a bottom-up perspective

Dataset-level

Most questions you want to ask involve many data points

typically somehow grouped in different ways

even for only moderately large datasets

In a way a top down approach to data

Mental model of a dataset

Marketing Vice President

How much did my new product generate
month by month, in the southern division, by user demographic, by sales
office, relative to the previous version, and compared to plan?

Marketing Manager

Give me sales statistics
by products, summarized by product categories, daily, weekly, and
monthly, by sale districts, by distribution channels.

Financial Controller

Show me expenses
listing actual vs budget, by months, quarters, and annual, by budget line
items, by district, division, summarized for the whole company.

Managers

can tell you what
measurement units are
important for them

think of the business in terms
of business dimensions

Mental model of a dataset

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Show me expenses listing actual vs budget, by months, quarters, and annual, by budget line items, by district, division, summarized for the whole company.



OLAP system design

End users usually can't fully describe what they want in a BI system:

- They can provide very important insight into how they think about the company:

- They can tell you what performance indicators are important to them

 - Each department can determine how it measures success in that particular department.*

- Give insight into how they combine and use different information to make strategic decisions

Multidimensional data



Multi-dimensional Model

For OLAP

Data is organized around subjects

Not geared towards a particular application; data in a data warehouse cut across applications

Data reflects states from the past

Data is rather frozen for analysis

States that can be measured by certain measures

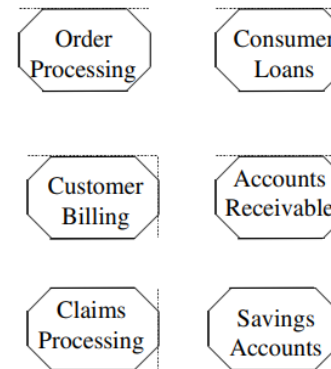
Measures can be further analysed – data crunching

Data is maintained for years

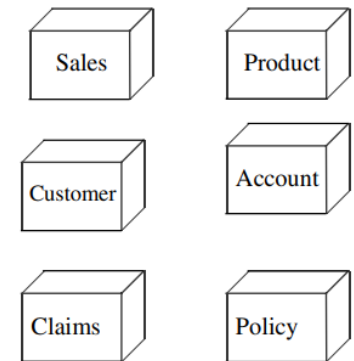
Broader time-horizon

Allow more generic analysis

Operational Applications



Data Warehouse Subjects



**Order Report Western Region
January 2009 (cont'd)**

Category	Product	SKU	Quantity Sold	Cost	Order Dollars
Packaging	Box - Large	011-4822	700	\$ 950.53	\$ 1,100.00
	Box - Medium	011-4899	1,250	\$ 1,001.84	\$ 1,380.00
	Box - Small	011-5744	1,200	\$ 1,200.72	\$ 1,330.00
	Clasp Letter	011-1729	400	\$ 352.82	\$ 356.00
	Envelope #10	021-0011	2,000	\$ 2,017.46	\$ 2,080.00
	Envelope Bubble	021-0012	1,200	\$ 866.51	\$ 1,212.00
	<i>All Packaging</i>			\$ 6,389.88	\$ 7,458.00
Pens	Gel Pen Black	017-1999	5000	\$ 116.39	\$ 120.00
	Gel Pen Blue	017-2444	2990	\$ 600.88	\$ 624.91
	Silver Pen	017-3001	50	\$ 128.46	\$ 130.00
	<i>All Pens</i>			\$ 845.73	\$ 874.91
Grand Total				\$207,229.42	\$214,896.91

Multi-dimensional Model



Measurement is easy to discern

whether by listening to people talk, or reading a report or a chart.



Every dimensional solution describes a process

by capturing what is measured.



Captured with information about different contexts

in which the measurements are evaluated.

Multi-dimensional Model

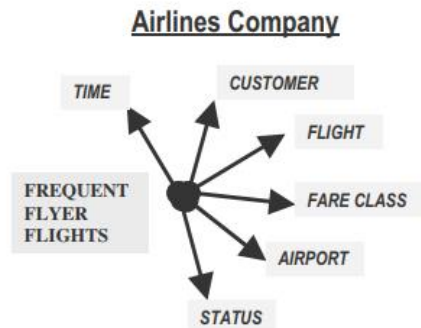
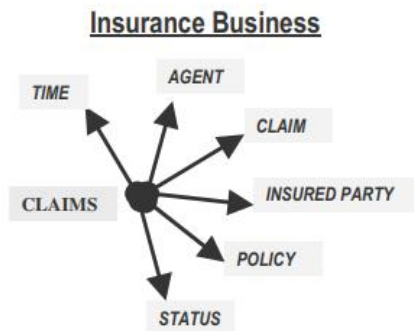
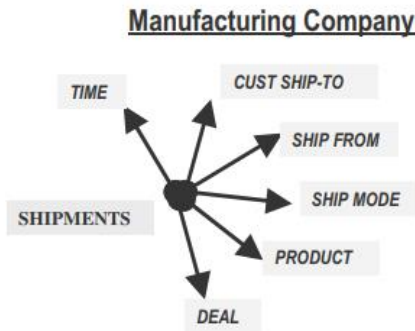
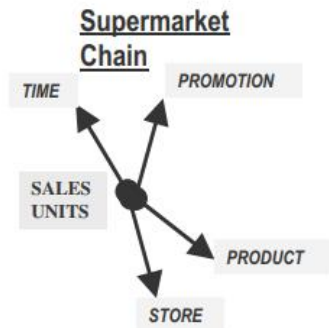
Managers

can tell you what measurement units are important for them

think of the business in terms of business dimensions

If your data warehouse users think in terms of measurements and business dimensions for decision making

you should also think of measurements and business dimensions



Managerial Perspective

	A	B	C	D	E	F	G
1	Row Labels	Internet Sales Amount	Reseller Sales Amount	KPI	KPI Goal	KPI Status	KPI Trend
2	2005	\$3,266,373.66	\$8,065,435.31	11331808.96	\$9,513,000.00	●	➡
3	2006	\$6,530,343.53	\$24,144,429.65	30674773.18	\$29,009,000.00	●	➡
4	2007	\$9,791,060.30	\$32,202,669.43	41993729.72	\$38,782,000.00	●	➡
5	2008	\$9,770,899.74	\$16,038,062.60	25808962.34	\$18,410,000.00	●	➡
6	2008 Q1	\$4,283,629.96	\$7,102,685.11	11386315.07	\$8,051,000.00	●	➡
7	2008 Jan	\$1,340,244.95	\$1,662,547.32	3002792.274	\$8,051,000.00	◆	➡
8	2008 Feb	\$1,462,479.83	\$2,700,766.80	4163246.633		●	➡
9	2008 Mar	\$1,480,905.18	\$2,739,370.98	4220276.163		●	➡
10	2008 Q2	\$5,436,429.15	\$8,935,377.49	14371806.64	\$10,359,000.00	●	➡
11	2008 Q3	\$50,840.63		50840.63		●	➡
12	2010					●	➡
13	Grand Total	\$29,358,677.22	\$80,450,596.98	109809274.2	\$95,714,000.00	●	➡

We sell products in various markets, and we measure our performance over time



Business Manager

Multi-dimensional Model



First there is
measurement

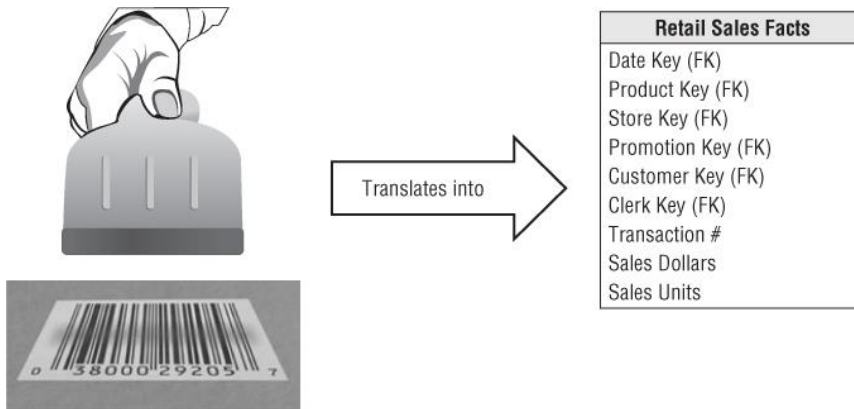
A series of measurements of
events/statuses dynamically
occurring in the reality



Measurement defines
performance metrics

Capture values of defined
performance metrics

Multi-dimensional Model



Facts are concepts of key importance for decision-making process

Correspond to events dynamically occurring in a company

The term fact represents business measurement

Imagine that you are standing in a store and watching the products being sold; unit quantity, sales amount, tax, etc. are recorded. for each product in each sales transaction.

These measurements are recorded when products are scanned at the checkout

Snapshots

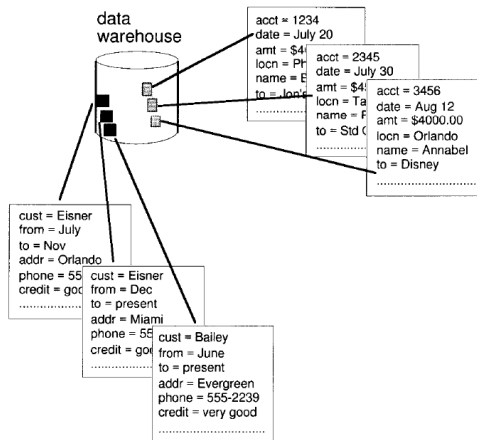
Data in the data warehouse is stored in units of "snapshots"

The records in the data warehouse are
created as of some moment in time
and are in effect a snapshot taken as of that
moment in time.

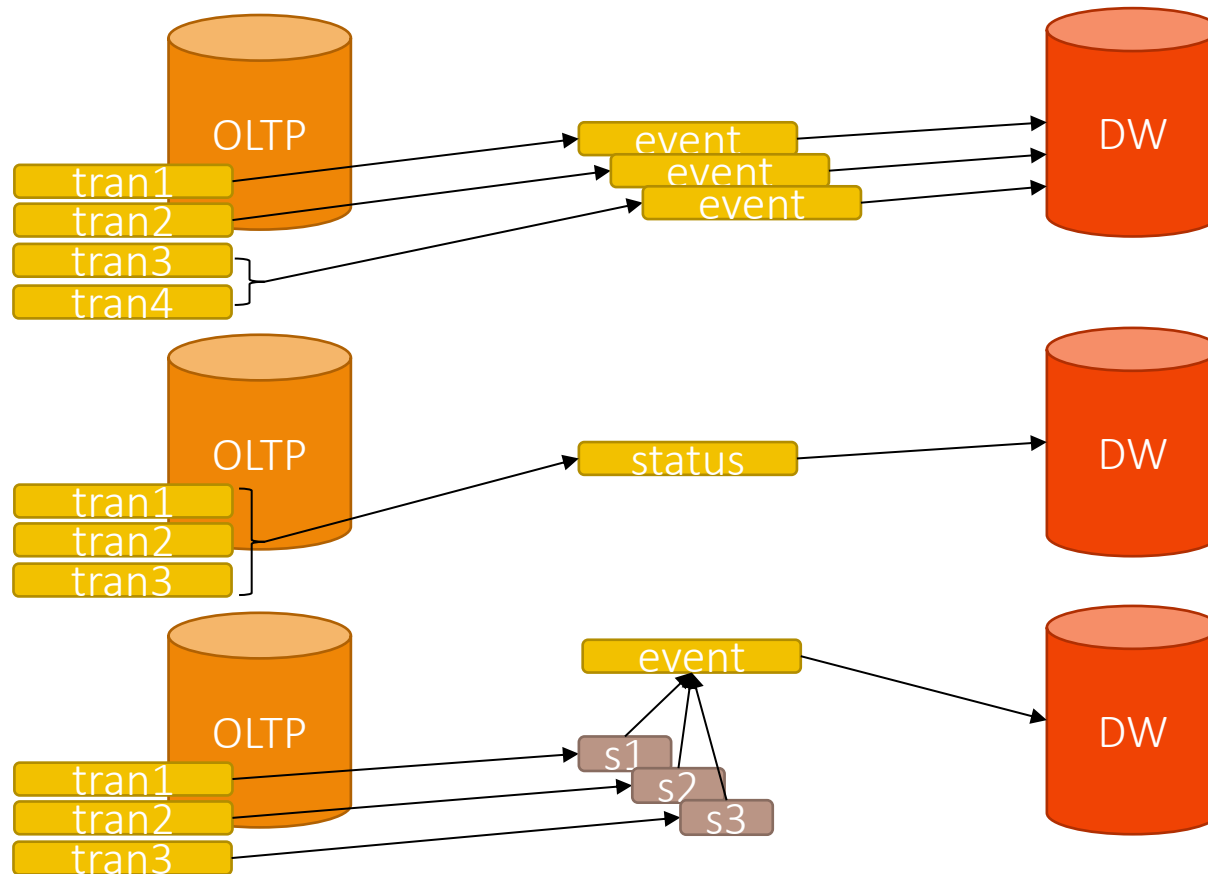
In this regard the data in the data warehouse is
fundamentally different from the data in an
operational data base environment.

Data in an operational data base environment can
be updated.

Since data in the data warehouse environment is
snapshot data it cannot be updated.



Facts



Multi-dimensional Model

Facts

In ER it can be represented as an Entity or n-ary relationship

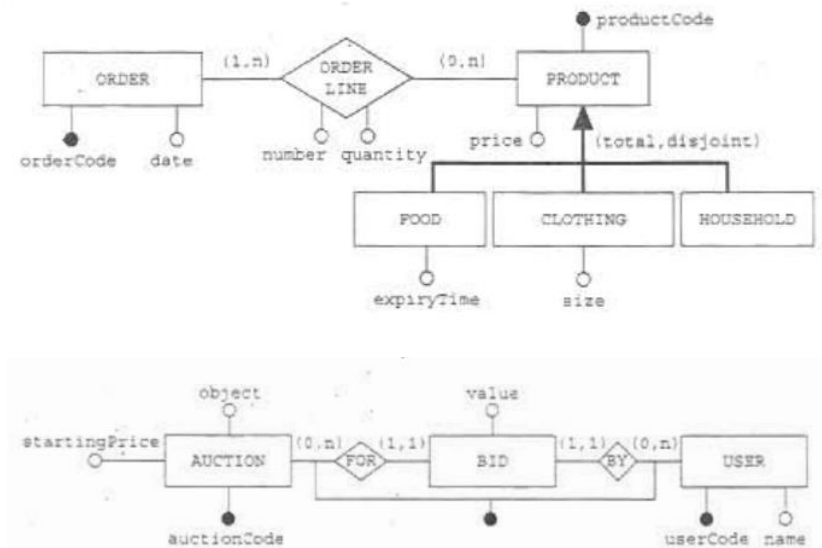
are the measurements that result from a business process event and are almost always numeric.

A fact corresponds to a physical observable event, and not to the demands of a particular report

defined at a certain level of detail, called grain, for example

a single product sold as part of a sales transaction

a daily summary of sales of individual products within daily sales transactions



Multi-dimensional Model



Measurements usually happens in a context

Directly or indirectly related to the measurement



Every dimensional solution describes the context

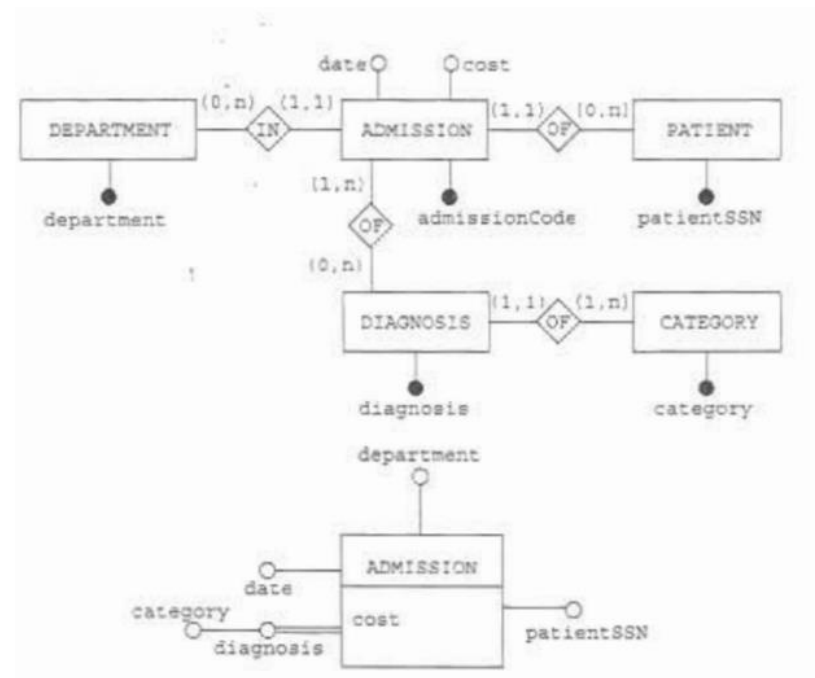
by capturing the context in which the measurements are evaluated.

Multi-dimensional Model

The concept of business dimensions is fundamental to the multidimensional model.

Business dimensions describe the business-specific objects within the model

such as products, customers, regions, employees, ...



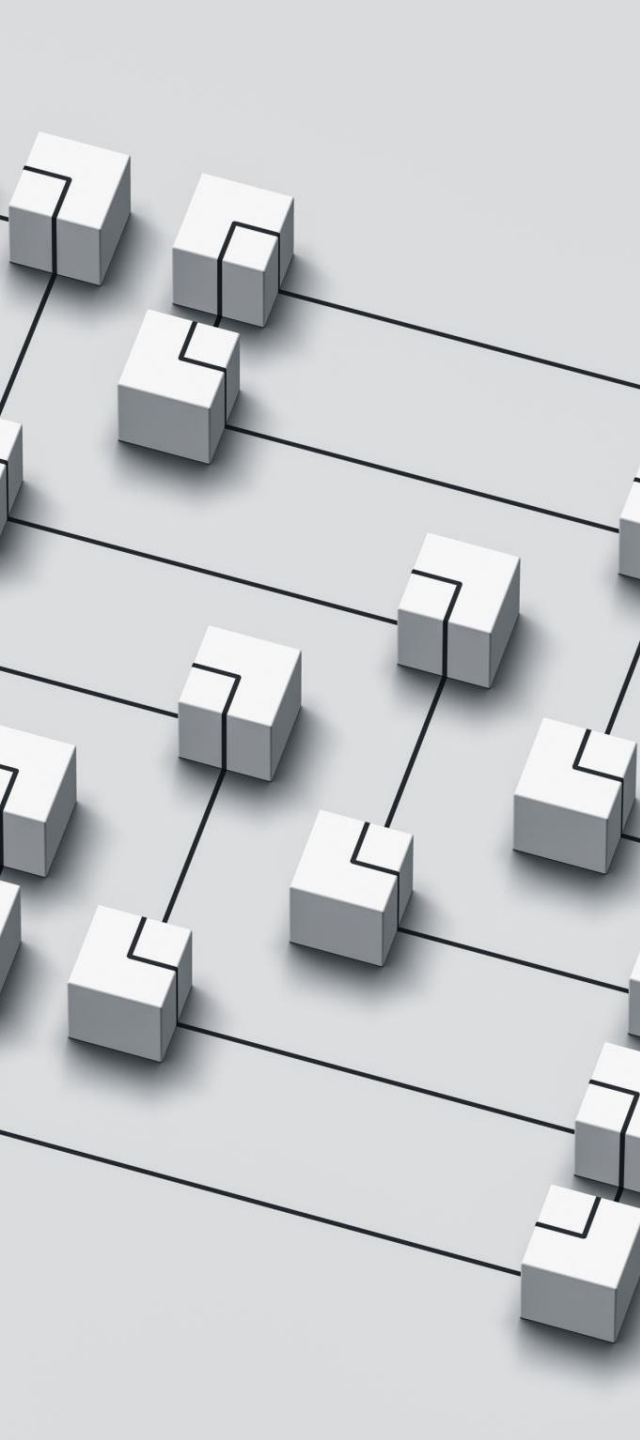
Multi-dimensional Model

Dimensions provide the “who, what, where, when, why, and how” context surrounding a business process event

- contain the descriptive attributes used by BI applications for filtering and grouping the facts

Dimension tables are sometimes called the “soul” of the data warehouse

- they contain the entry points and descriptive labels that enable the DW/BI system to be leveraged for business analysis



Multi-dimensional Model

These two simple concepts, **measurement** and **context**, are the foundation of dimensional design.

A multidimensional data model is organized around a central theme

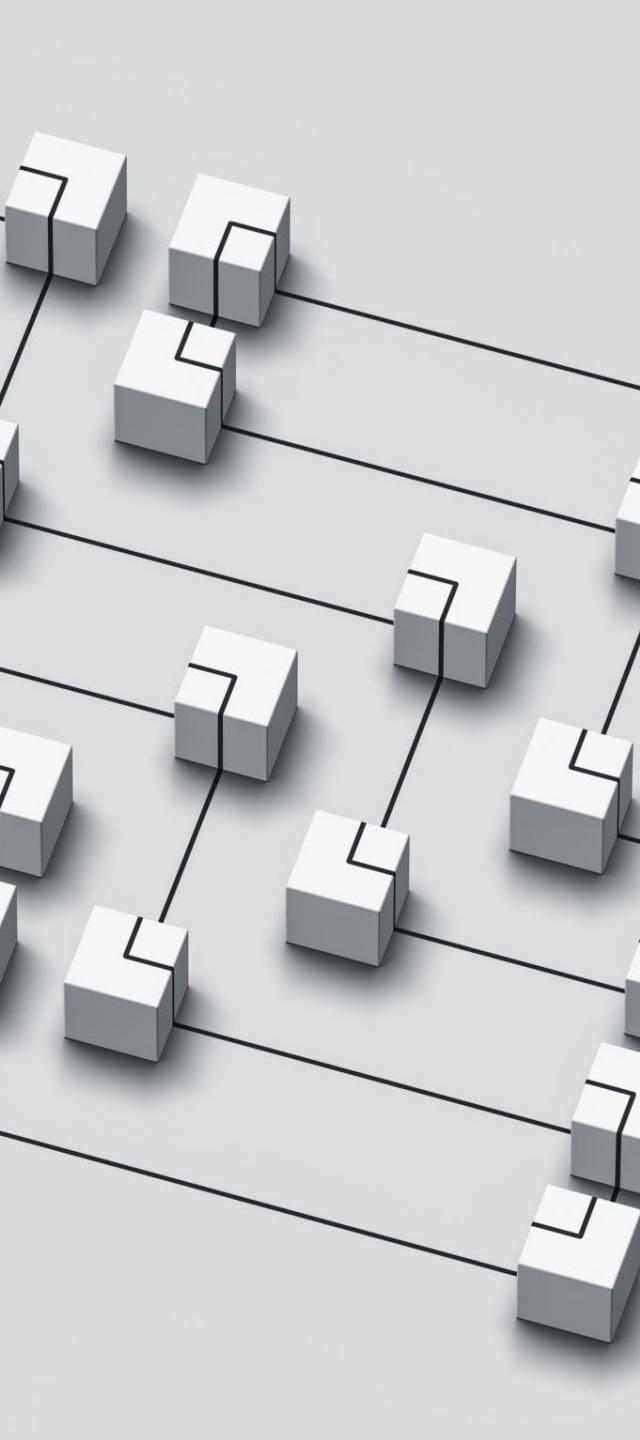
- ability to define a data by subject matter, for example – sales

- delivers information about a theme, rather than the organisation's current operations

Every dimensional solution describes a process by capturing what is measured and the context in which the measurements are evaluated.

- measurements are called **facts**

- context descriptors are called **dimensions**



Multi-dimensional Model

Facts and measures

Phenomena we are focused on (Facts), and the numerical values that describe the phenomena (Measures)

Eg. Each sales record is a fact, and its sales value is a measure

central element of the dimensional model and further data warehouse

Dimensions

Perspectives on the phenomena

Group correlated attributes into the same dimension to ease analysis tasks

Eg. Each sales record is associated with its values of Product, Store, Time

In general

if an attribute is commonly aggregated or summarized, it is a fact

if an attribute is used to drive aggregations or summarizations, it is a dimension.

Let us quickly review the business model of a large retail operation.

The daily sales are meaningful only when they are related to the dates of the sales, the products, the distribution channels, the stores, the sales territories, the promotions, and a few more dimensions.

Very few models are limited to three dimensions or less.

For example, they compare actual sales against targets and against sales in prior periods.

Facts and Dimensions

Fact:

A fact is something that has happened or been measured.

Like sale of a single product to one customer or the total amount of sales of a particular item in a month.

fact is a numerical value

usually associated with several dimensions

Dimension:

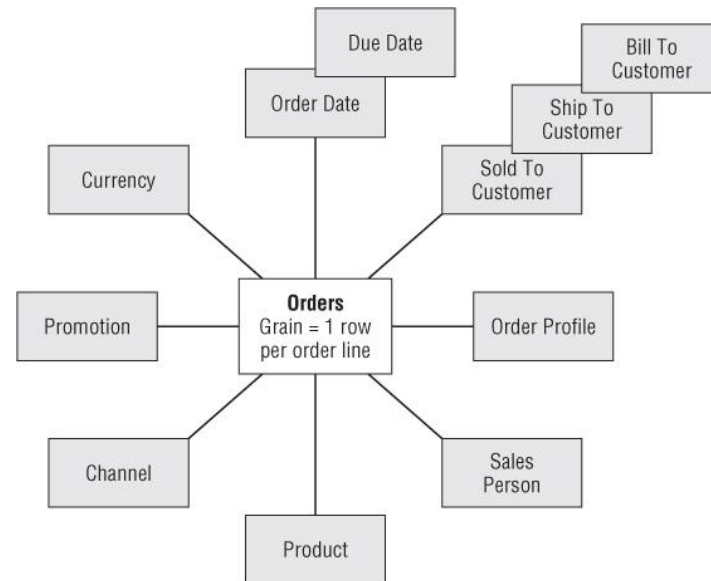
A dimension is the main analytical object in a BI space.

can be a list of products or customers, time, geographic location, or any other entity that is used to analyze numeric data.

Dimensions are related to facts.

The reason for their existence is the addition of qualitative information to the numerical information contained in the facts.

Sometimes a dimension may have to do with other dimensions, but directly or indirectly it will always be somehow related to the facts.



Multi-dimensional Model

Query

- Show me sales, profit and average call volume per day for my 10 most profitable salespeople

Components

- Facts
 - *a focus of interest for decision-making - subject*
- Measures
 - *attributes that describe facts from different points of view - value*
- Dimensions
 - *discrete attributes which determine the granularity adopted to represent facts - perspective*



Multi-dimensional Model

To sum up

A multidimensional data model is organized around a central theme

ability to define a data by subject matter

for example – sales

delivers information about that theme (e.g., sales)

rather than the organisation's current operations –
handling of particular sales order

focuses on snapshots – measurements

of the operational data at given moments/events

Analytical system

Interaction with an analytic system takes place exclusively through queries that retrieve data about business processes;

information is not created or modified

queries involve large numbers of transactions,

in contrast to the operational system's typical focus on individual transactions.

specific questions asked are less predictable,

and more likely to change over time.

historic data will remain important to the analytic system

long after its operational use has passed.

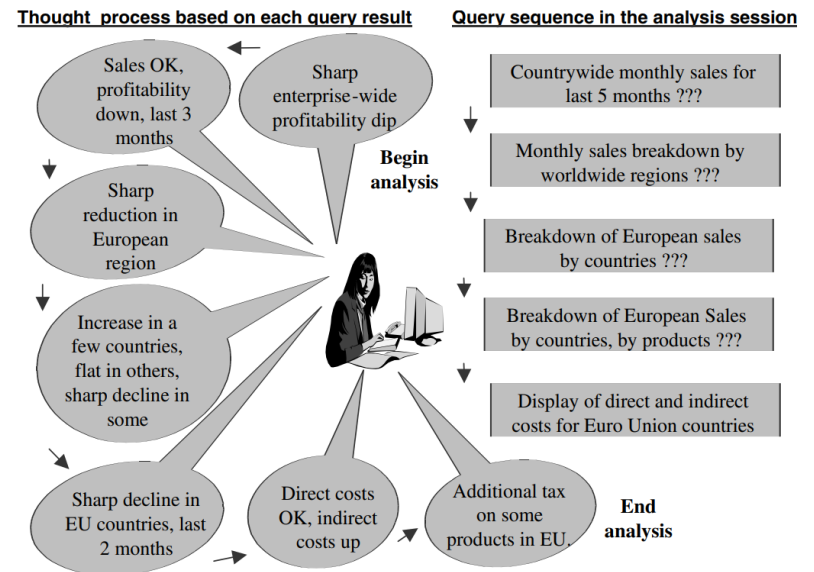


Analytical system

The analysis does not stop with this single multidimensional query.

The user continues to ask further questions

*comparisons to similar products,
comparisons among territories,
views of the results by rotating the
presentation between columns and
rows*



Analytical system

Differs from normalised data model in the degree of data normalisation

Limited elasticity

It is relatively easy to add new tables to a normalised model

Higher risk of data inconsistency

Data redundancy

Storing calculated values

Efficiency

Intuitiveness

OLAP

Common types of OLAP needs:

Comparisons (this period v.s. last period)

Show me the sales per store for this year and compare it to that of the previous year to identify discrepancies

Ranking and statistical profiles (top N/bottom N)

Show me sales, profit and average call volume per day for my 10 most profitable salespeople

Custom consolidation (market segments, ad hoc groups)

Show me an abbreviated income statement by quarter for the last four quarters for my northeast region operations

Managerial Perspective

Consider an example of a car rental company

Facts ?

Measures ?

Dimensions ?

Consider an example of a medical center

Facts ?

Measures ?

Dimensions ?



DATA as a cube

A 3D cube composed of smaller colored cubes, representing data as a cube. The cube is made of 27 smaller cubes arranged in a 3x3x3 grid. The colors of the visible faces include orange, green, yellow, red, blue, and white. The text "DATA as a cube" is overlaid in white.

Managerial Perspective

Data Warehouse users
That translate directly to :

Facts and measures:

Sales amount, Sales value, Profit, etc.

And dimensions:

Product, Markets, Time, etc.

We sell products in various markets, and we measure our performance over time



Business Manager



We sell *Products* in various *Markets*, and we measure our performance over *Time*

Data Warehouse Designer



Data as a cube

Logical

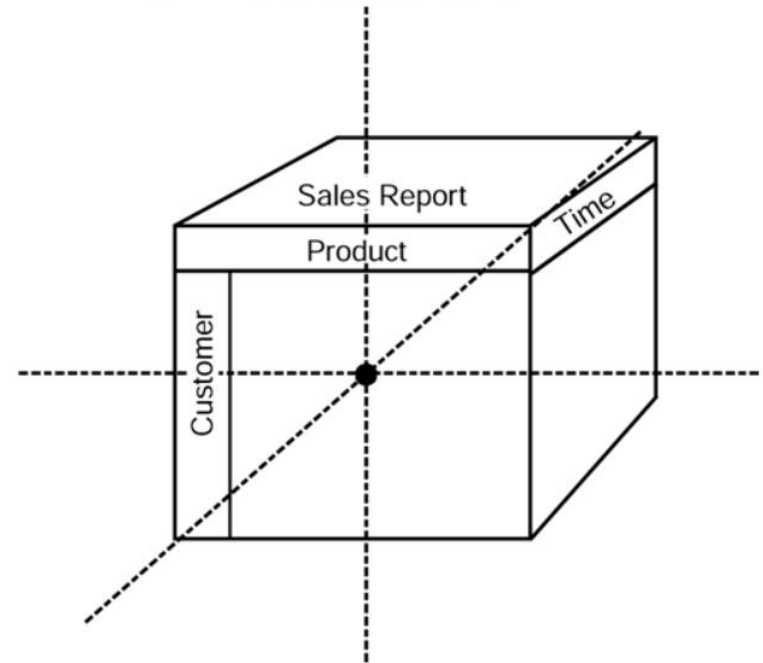
data as a multidimensional cube

Facts and dimensions form a cube

Axis represent dimensions

Points represent measures

Figure 5.2. Example of a three-dimensional cube.



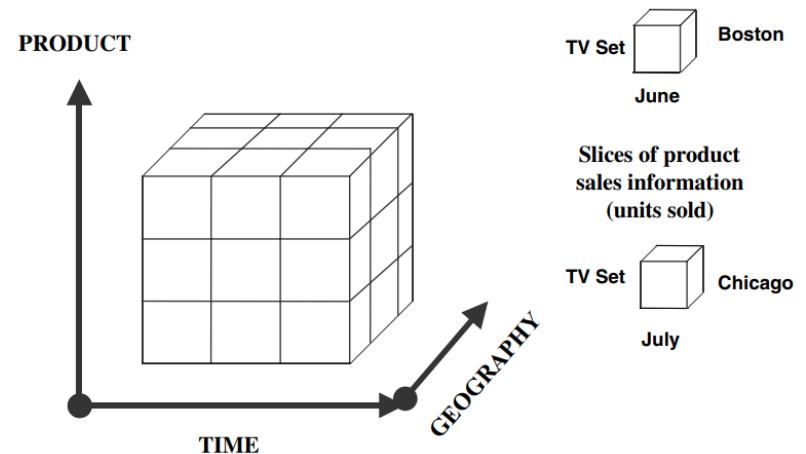
Data as a cube

Analysis of sales units along the three business dimensions of product, time, and geography

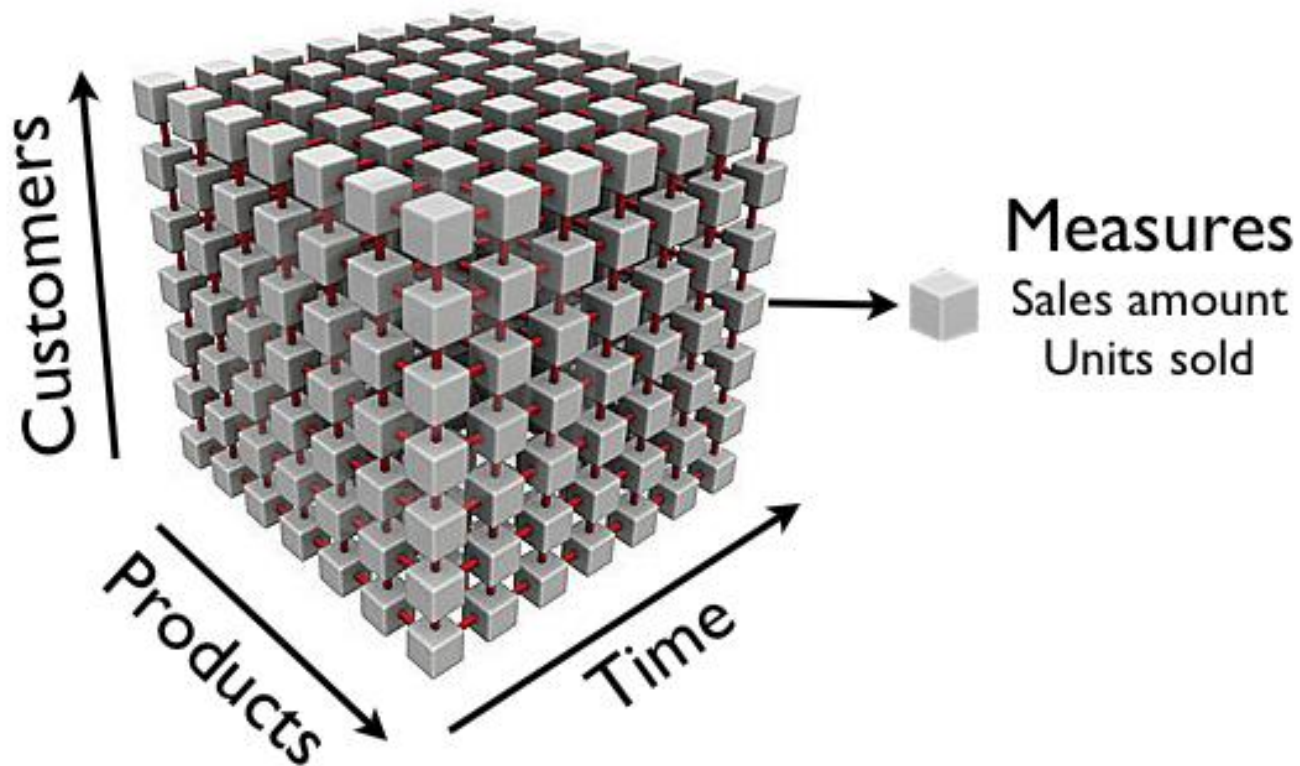
Three dimensions form a collection of cubes

In each of the small dimensional cubes

you find the sales units for that particular slice of time, product, and geographical division



Data as a cube



Data as a cube

Multi-dimensional data

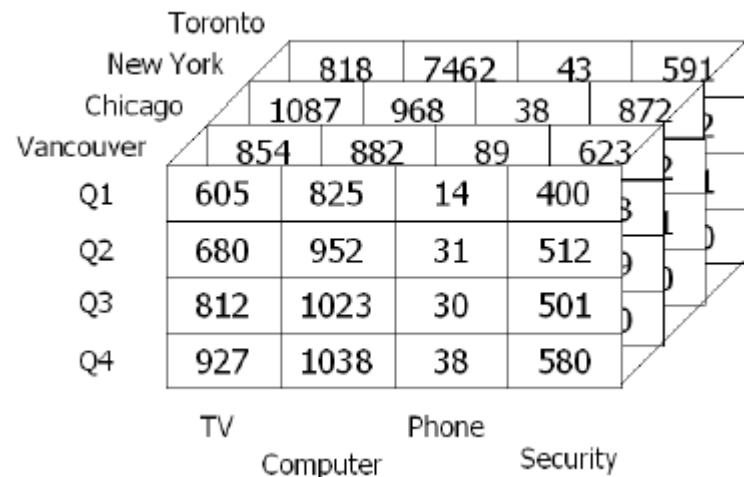
Each measure's value is accessed

Using coordinate system composed of a set of dimensions

Particular attributes of the dimensions define a single point in the cube

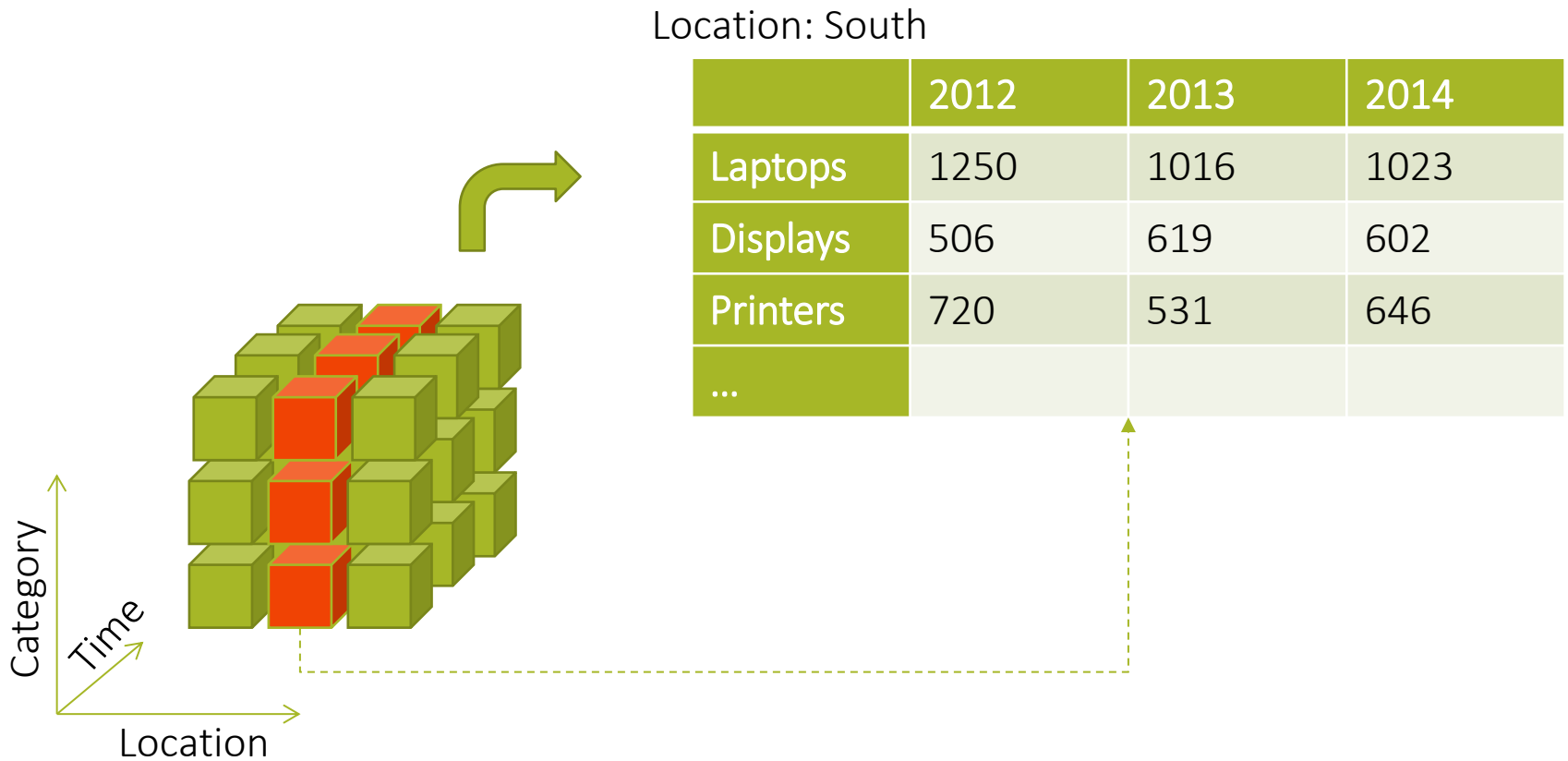
Example

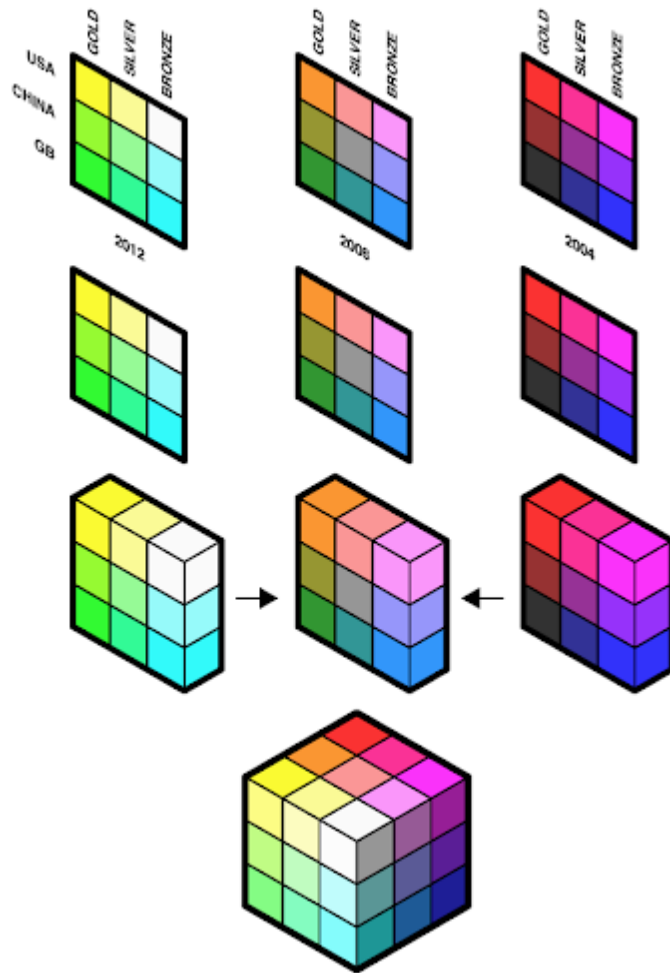
Q1, Vancouver, Computer → 825



	Toronto	New York	Chicago	Vancouver
Q1	605	825	14	400
Q2	680	952	31	512
Q3	812	1023	30	501
Q4	927	1038	38	580
	TV	Computer	Phone	Security

Data as a cube

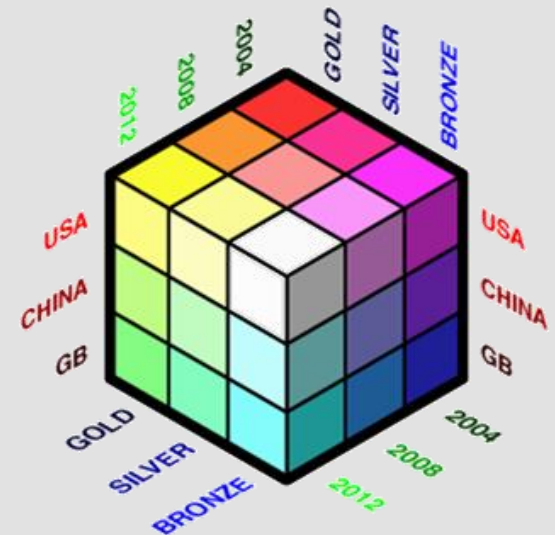




Data as a Cube

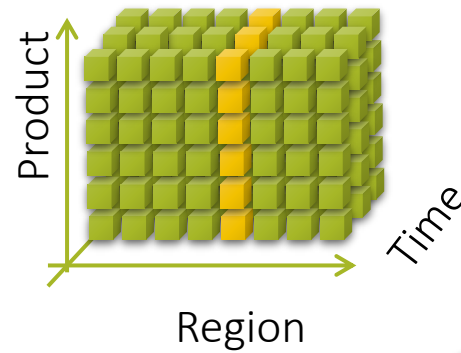
Multidimensional

many dimensions, many
axeses

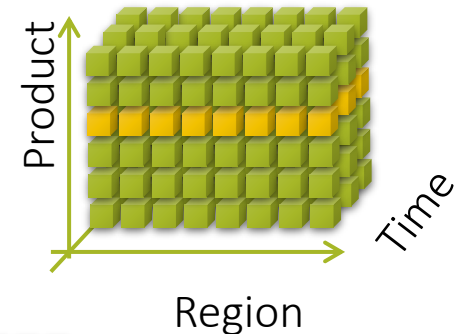


Data as a cube

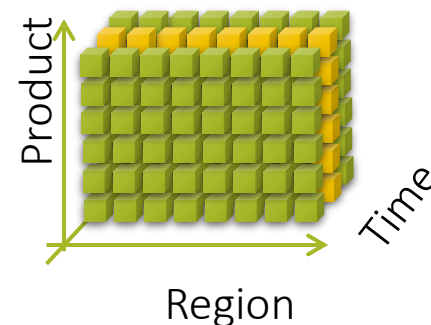
Multi-dimensional data
Regional manager perspective
Dimension



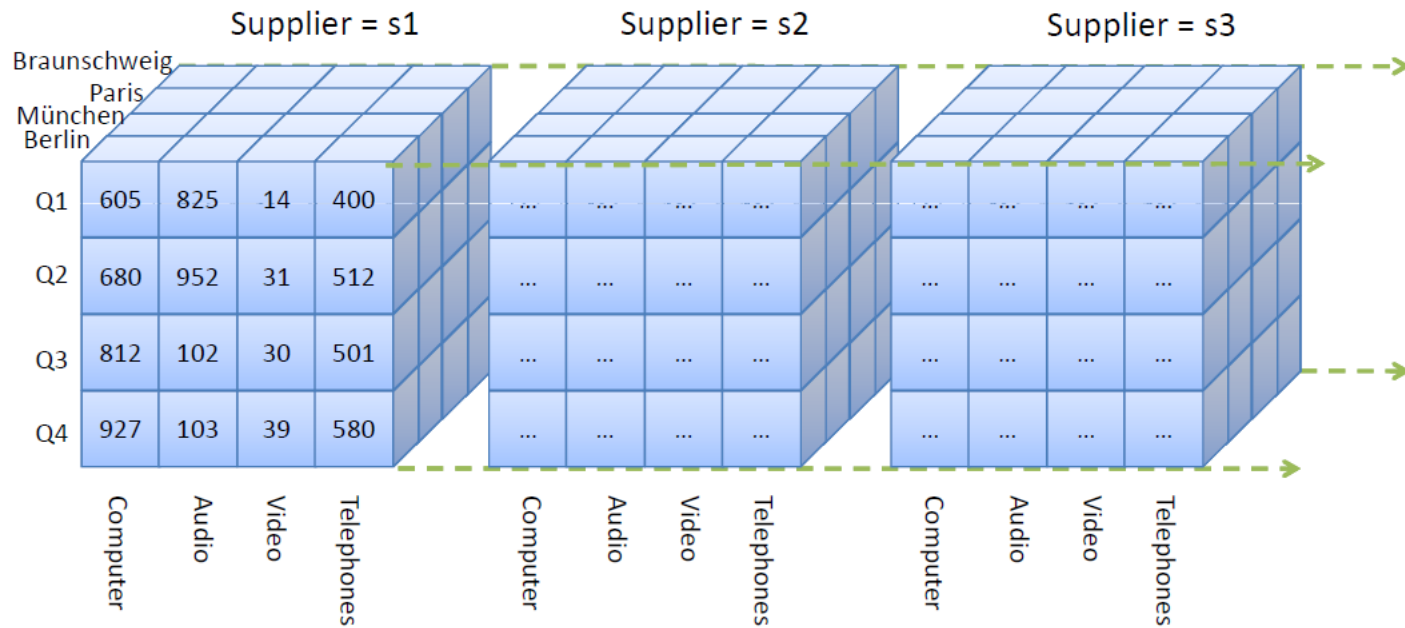
Multi-dimensional data
Product manager perspective
Dimension



Multi-dimensional data
Financial manager perspective
Dimension



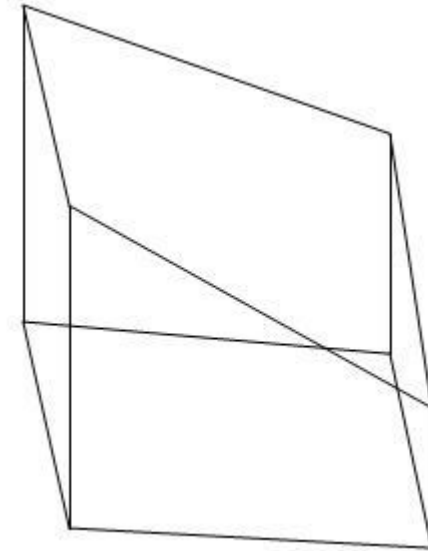
Data as a cube



Data as a cube

Contrary to what the name suggests

a data cube /OLAP
cube/hypercube is not necessarily
a strict mathematical cube
as its sides are not always equal.



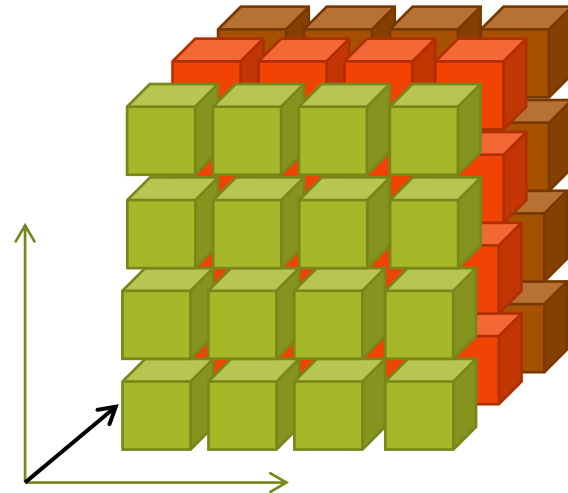
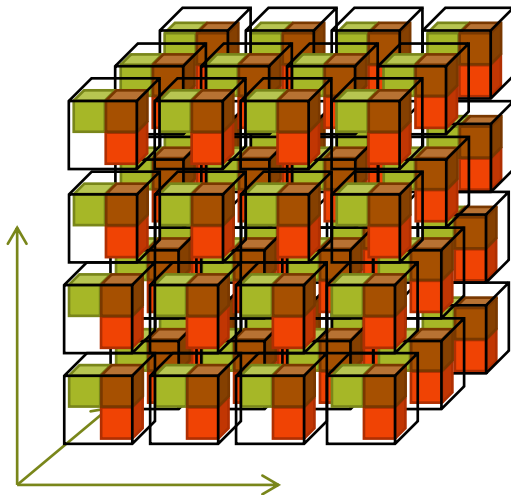
Data as a cube

Multi-dimensional data

Multiple measures

A point in a multi-dimensional space can represent several values

Add virtual dimensions representing measures



Data as a cube

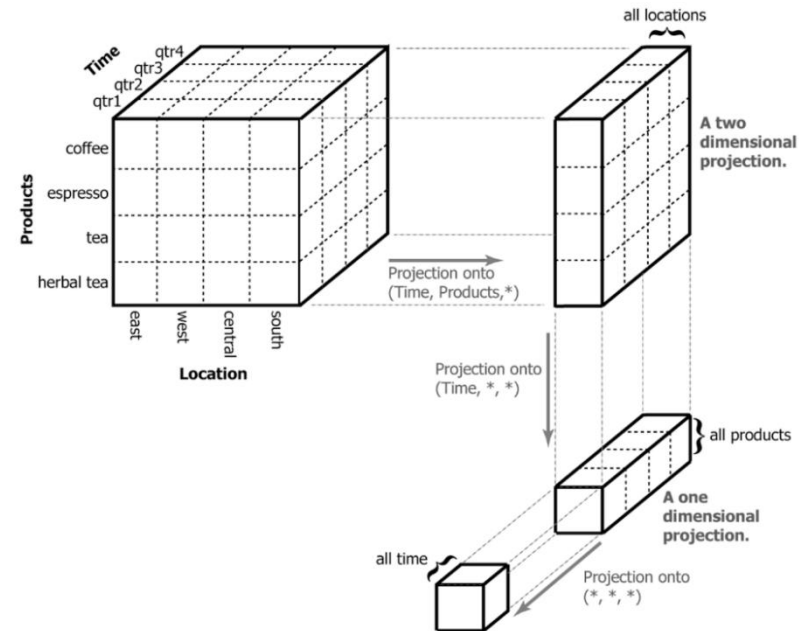
Left to right

How fine-grained data turns into a cube.

Right to left

How the cube containing all the data in one value is broken down:

first by time, then by product, then by location.



Data as a cube

How do you ask questions of such a dataset?

You specify how you want it broken down and how to aggregate the measures.

Example

Dataset of incomes for many different job titles, ages, genders, education levels, etc.

Ask about the average income for men vs. women

It's the dataset perspective

Process:

Split the dataset along the gender dimension – into two groups.

Calculate the averages for each gender.

Data as a cube

This way, you can quickly ask many different questions:

What's the average by education level?

How about gender *and* education level?

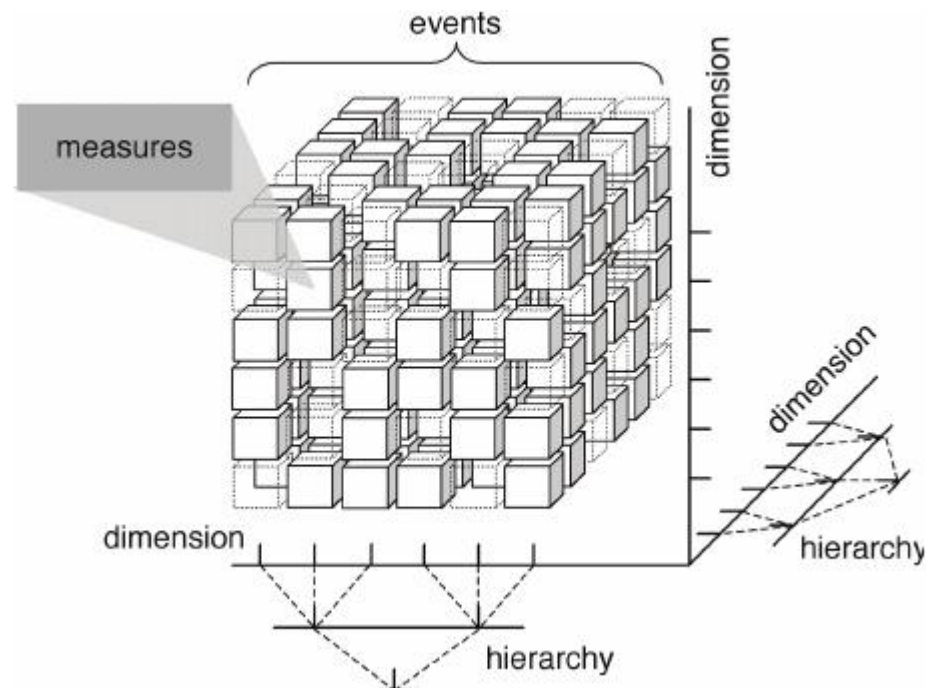
What about age?

Etc.

Different aggregations in each slice are possible

average, median, minimum, maximum, etc.

It's all just a matter of asking the question.





Why to model?

A lot of different data sources needed to meet the needs of users

Purely engineering approach is hard

very often you are just overwhelmed with information sources, without any idea on how to handle and organise them

The right data modelling strategy is needed

Various user needs

More than ever different types of data consumers

with different types of expectations on what they want to see out of their data and how to use it

Optimization

Managing speed and costs

Clarity

Provides a clear strategy on how to work and organise data

Multi-dimensional Model

A design paradigm

- Logical representation of data

- Focused on data aggregation

Is a widely accepted

- the preferred technique for presenting analytical data

- it meets two simultaneous requirements:

 - Deliver data that business users understand.*

 - Allow for efficient query performance.*



Data Model

Conceptual and abstract

Does not specify the details of the physical implementation

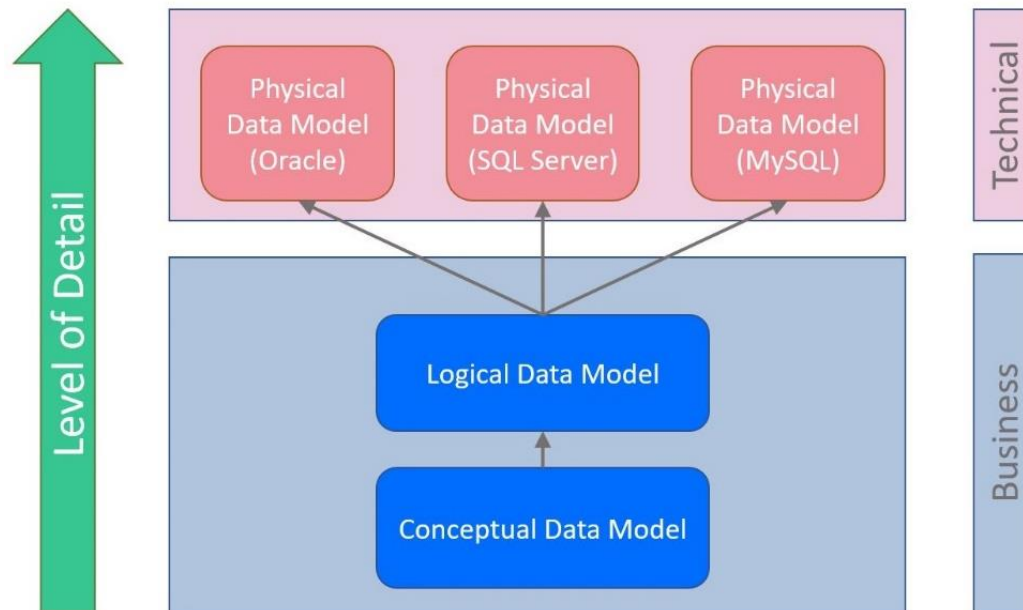
Defines the types of information needed

Three objectives:

Simplicity

Completeness

Efficiency



Multi-dimensional Model

Product	Type	Category	Store	City	County	Date	Sales
Top	Beer	Beverage	Trøjborg	Århus	Århus	25 May, 2009	5.75

Facts and measures

Facts

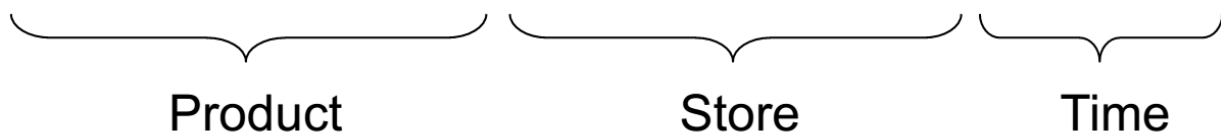
describe business processes, certain business events

Measures

quantitative context

aggregated by BI system to establish summary information

Product	Type	Category	Store	City	County	Date	Sales
Top	Beer	Beverage	Trøjborg	Århus	Århus	25 May, 2009	5.75



Dimensions

Dimensions

provide the “who, what, where, when, why, and how” context
surrounding the subject – business process event

contain the descriptive attributes

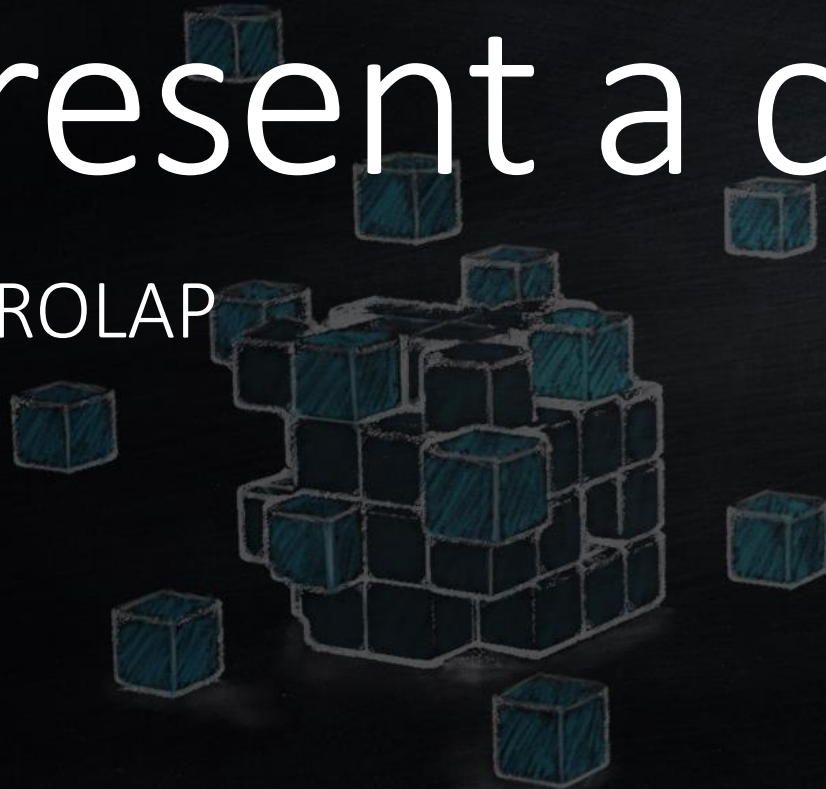
used by BI applications for filtering, ordering and grouping of the facts

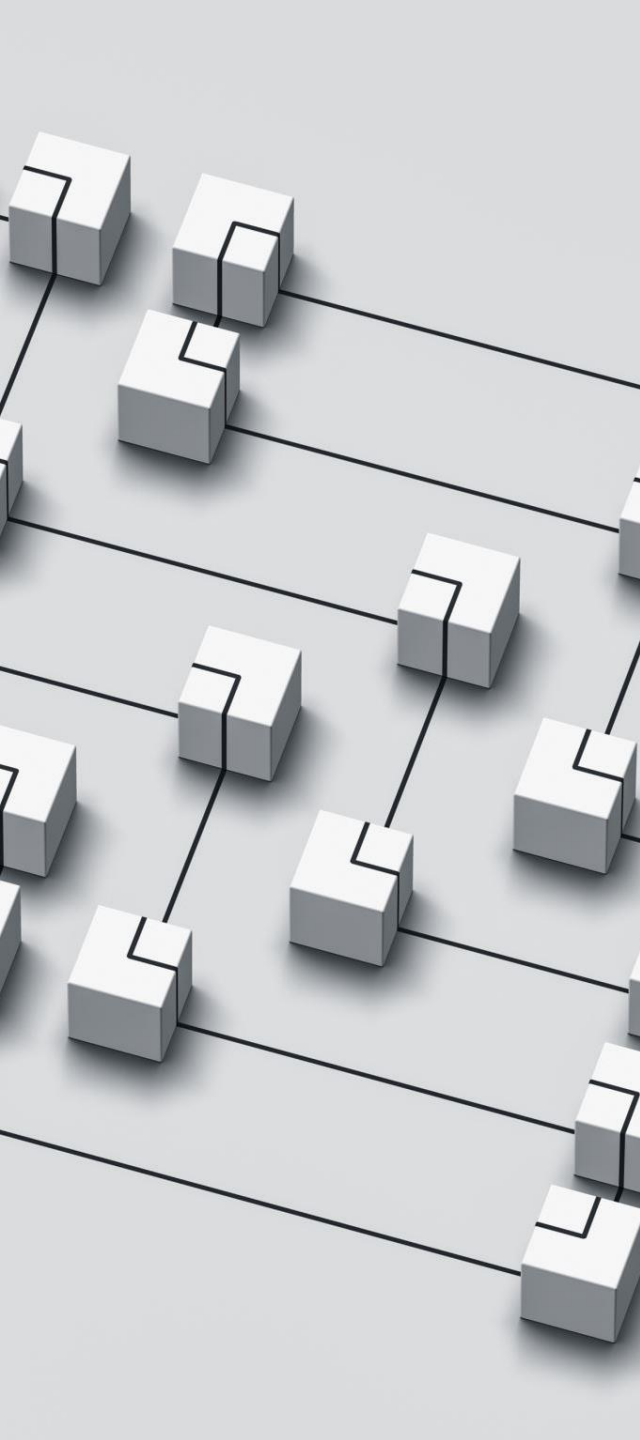
Product	Type	Category	Store	City	County	Date	Sales
Top	Beer	Beverage	Trøjborg	Århus	Århus	25 May, 2009	5.75

Product Store Time

How to physically represent a cube?

MOLAP / ROLAP





Data as a cube

Implemented in a relational database,

the dimensional model is called a star schema.

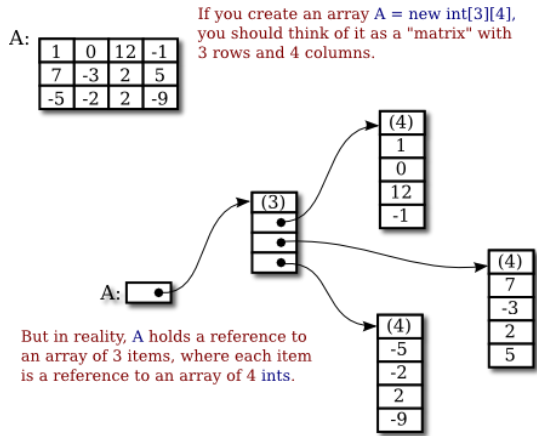
Implemented in a multidimensional database,

it is known as a data cube.

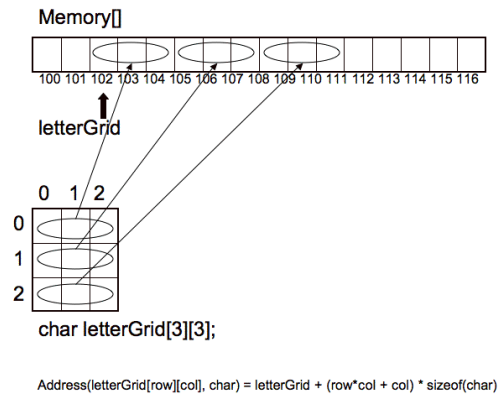
If any part of your data warehouse includes a star schema or a cube

it leverages dimensional design.

Data as a data cube

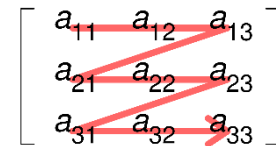


JAVA

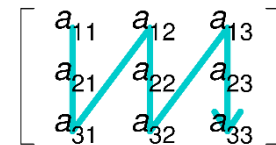


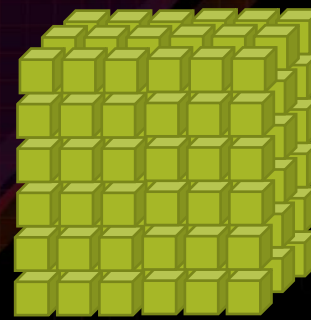
C like

Row-major order



Column-major order





DW Schemas

a reporting schema is employed that is optimized for reads and aggregations.

Cubes and tables

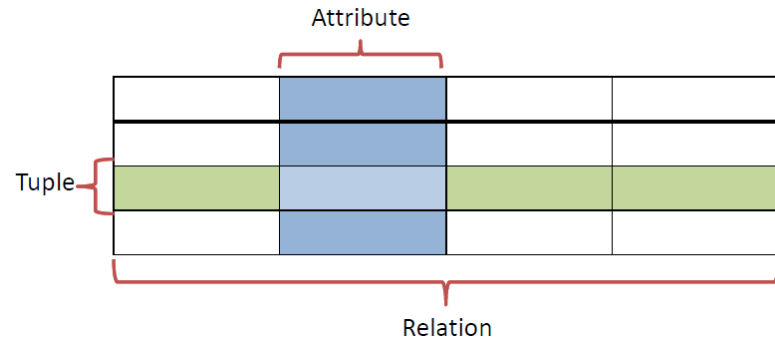
DB in relational model

the content is modeled as a set of relations

a relation is defined as a set of tuples that have the same attributes

is usually described as a table

each tuple is identified by a distinctive unique key



3NF

Third Normal Form (3NF) design

seeks to minimize data redundancy and avoid anomalies in data insertion, updates and deletion.

has a long heritage in online transaction processing (OLTP) systems.

OLTP systems must maximize performance and accuracy when inserting, updating and deleting data.

Transactions must be handled as quickly as possible or the business may be unable to handle the flow of events, perhaps losing sales or incurring other costs.

avoid redundant data manipulation and minimize table locks

both of which can slow inserts, updates and deletes.

3NF in Data Warehousing

3NF data model is a foundational approach in data warehousing

For structuring integrated, consistent, and redundancy-free data

Minimized Redundancy:

3NF eliminates duplicate data by splitting tables into smaller, related entities.

Each non-key attribute depends directly on the primary key, ensuring data is stored only once

Data Integrity:

Enforces referential integrity through primary/foreign key relationships, reducing anomalies during updates or deletions

Subject Orientation:

Groups data by business subjects (e.g., "Customer," "Product") rather than source systems, enabling a unified view across the organization

3NF in Data Warehousing

3NF designs have great flexibility, but it comes at a cost:

Query Complexity:

Analytical queries often require joins across multiple tables, leading to slower performance

compared to denormalized models

Complex Design:

Achieving 3NF requires meticulous upfront modeling

especially for large enterprises

ETL Complexity:

Transforming raw data into 3NF demands sophisticated ETL pipelines to handle business rules and source system discrepancies

User Accessibility:

Not optimized for direct business user analysis

often requires a separate presentation layer for BI tools

Data Cubes

Data Cube in RDB

Dimension tables

Descriptive attributes

eg. item (item_name, brand, type) or time(day, week, month, quarter, year)

holds different perspectives on the data

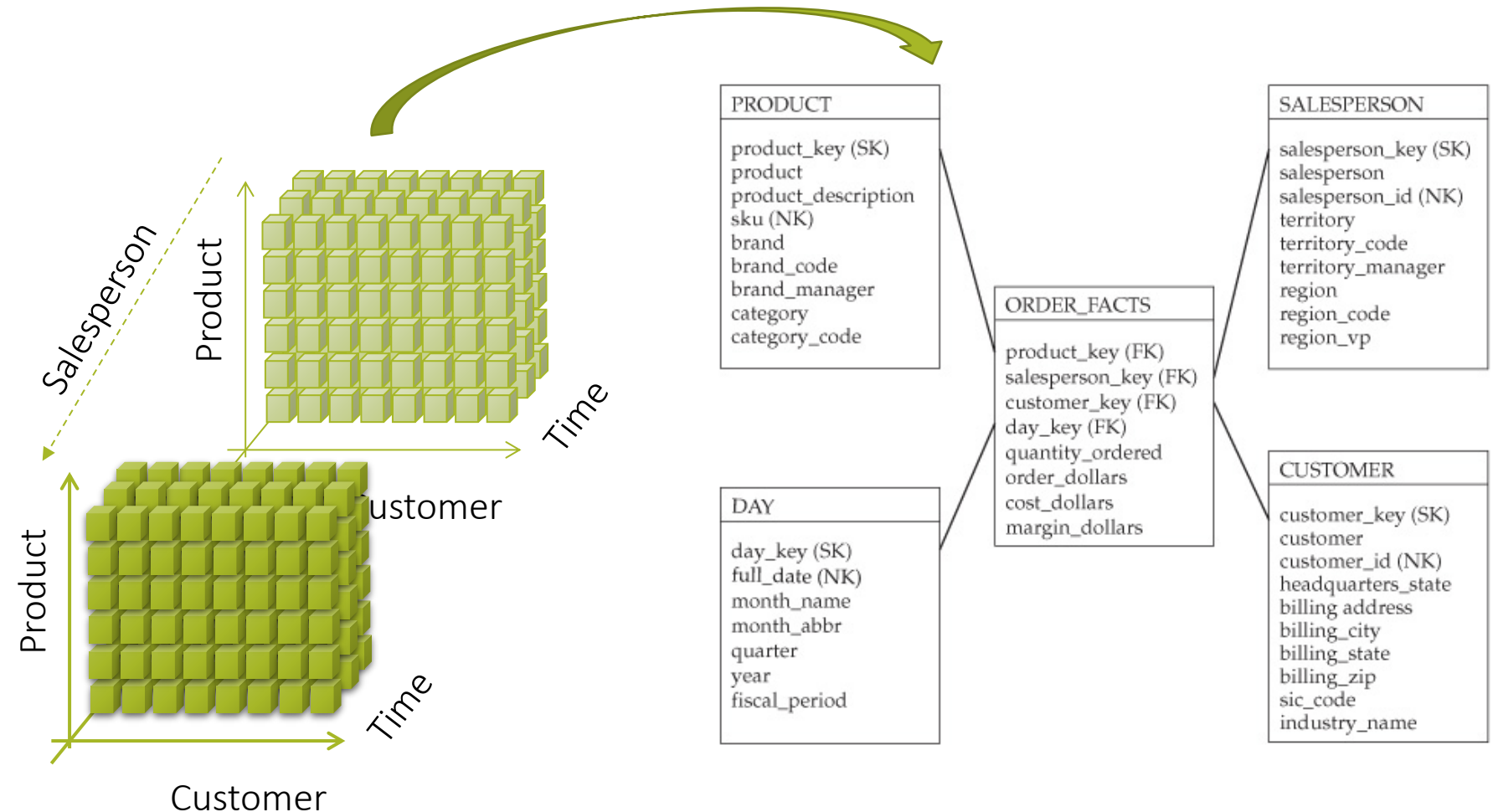
Fact table

contains measures

eg. dollars_sold, amount sold

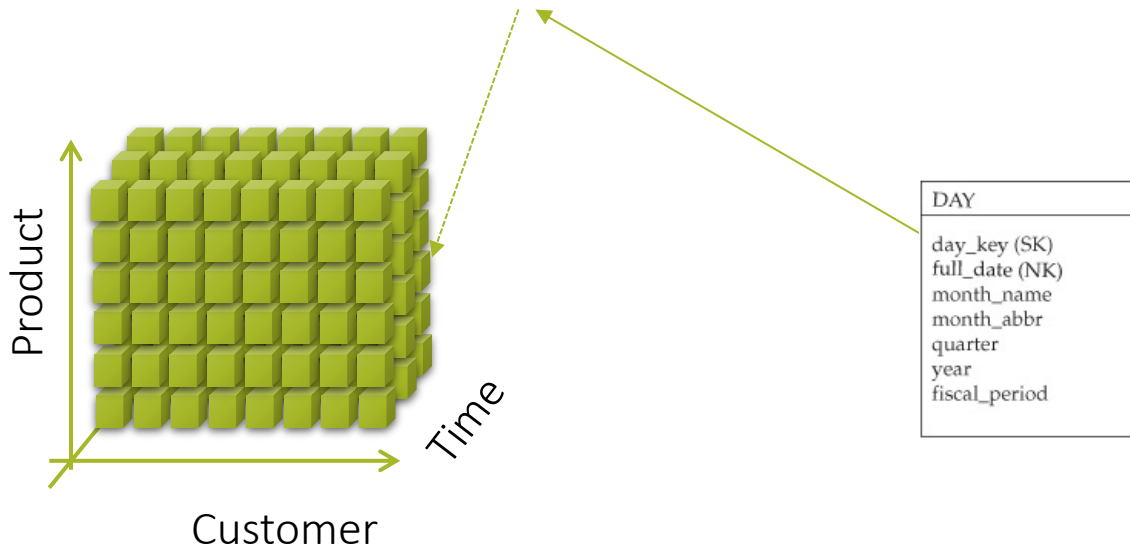
contains keys to each of the related dimension tables

Cubes and Stars



Cubes and Stars

product_key	day_key	salesperson_key	customer_key	quantity_ ordered	order_ dollars	cost_ dollars
102291	3831	2991	240123	882	8822.29	8028.28
194482	2931	1992	572339	249	2499.29	2274.35
183882	2983	2933	937759	3394	33940.29	30885.66
102291	3831	2991	240123	882	8822.29	8028.28



Data warehouse schema

Basic conceptual schemas:

Star schema

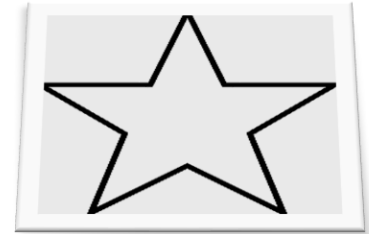
the most popular way to organize a reporting schema

Snowflake schema

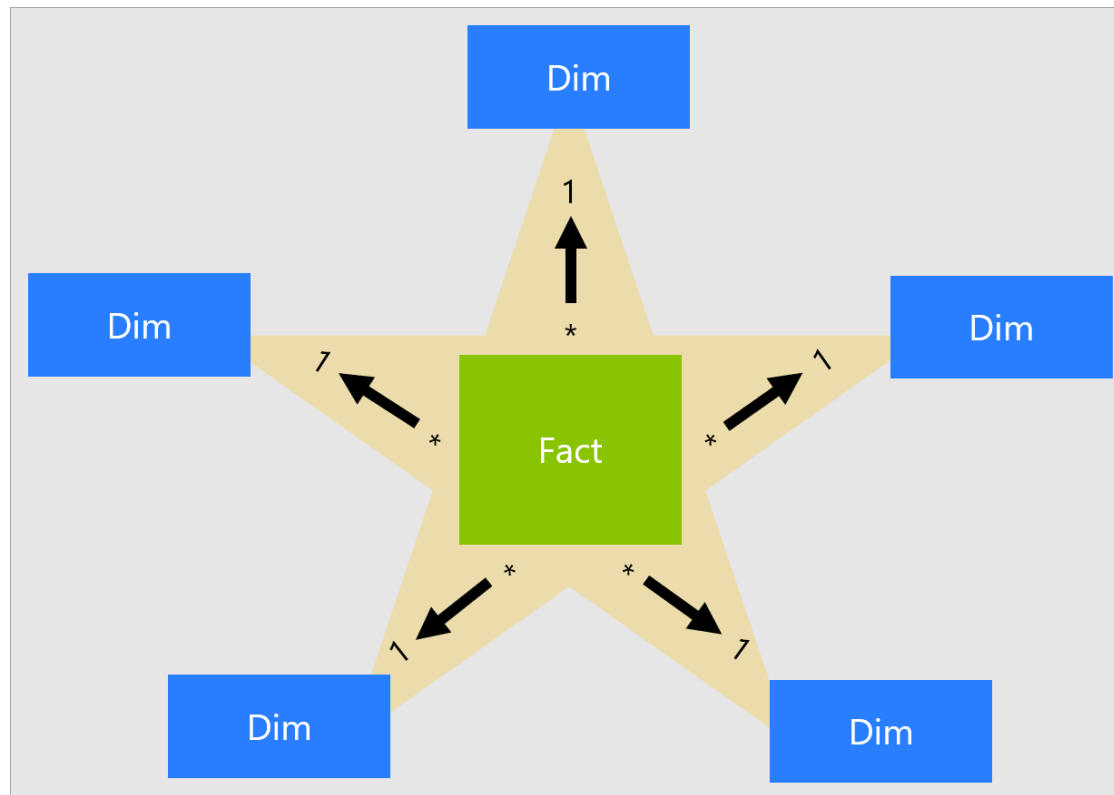
when star is not enough

Fact-constellation schema

when there are multiple facts



Star Schema



Star Schema

Star schema:

- most data warehouses adopt a star schema to represent the multidimensional model

- each dimension is represented by a dimension table

 - eg. LOCATION (location_key, street, city, state, country, region)

 - dimension tables are not normalized*

 - is built for simplicity and speed*

- snapshots are described through a fact-table

 - each tuple consists of*

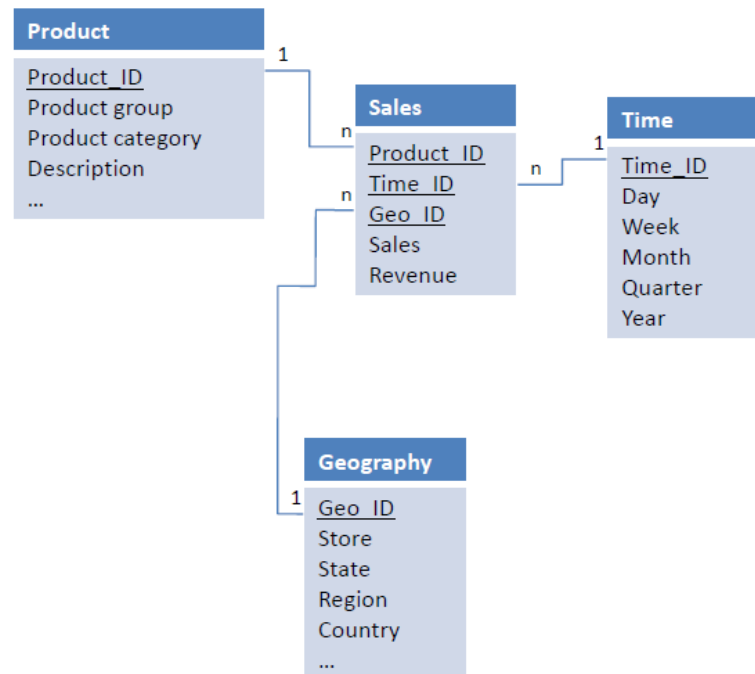
 - a pointer to each of the dimension-tables - (foreign-key)

 - a list of measures (e.g. sales amount, sales value, etc)

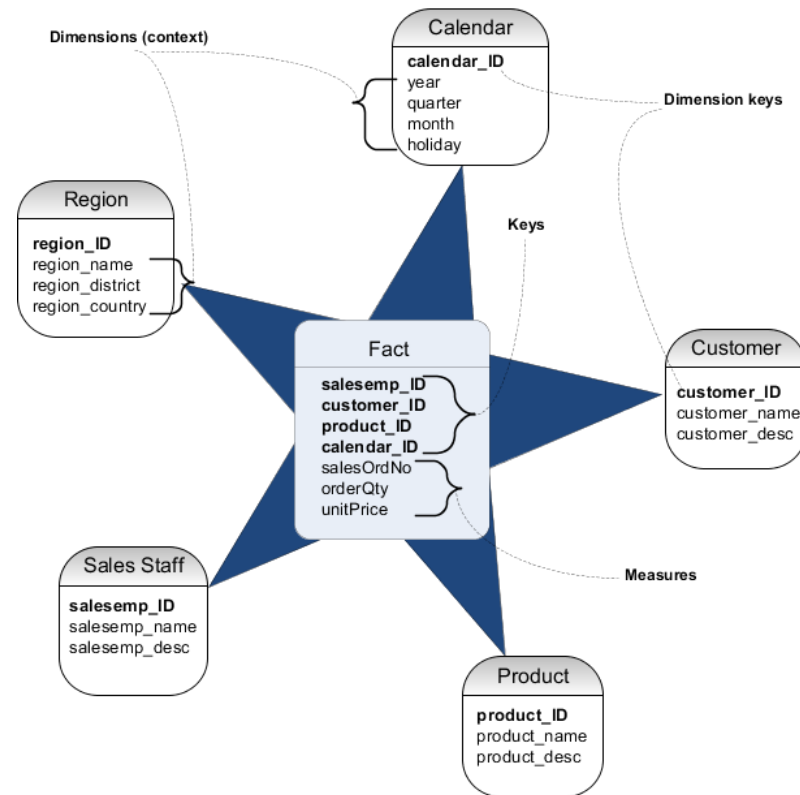
Star Schema

Star schema:

A fact table in the middle connected to a set of dimension tables

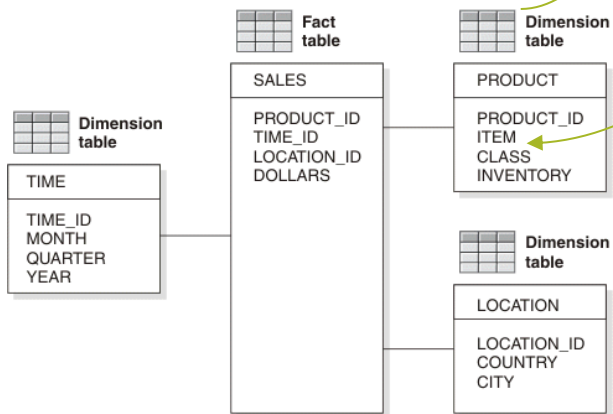


Star Schema

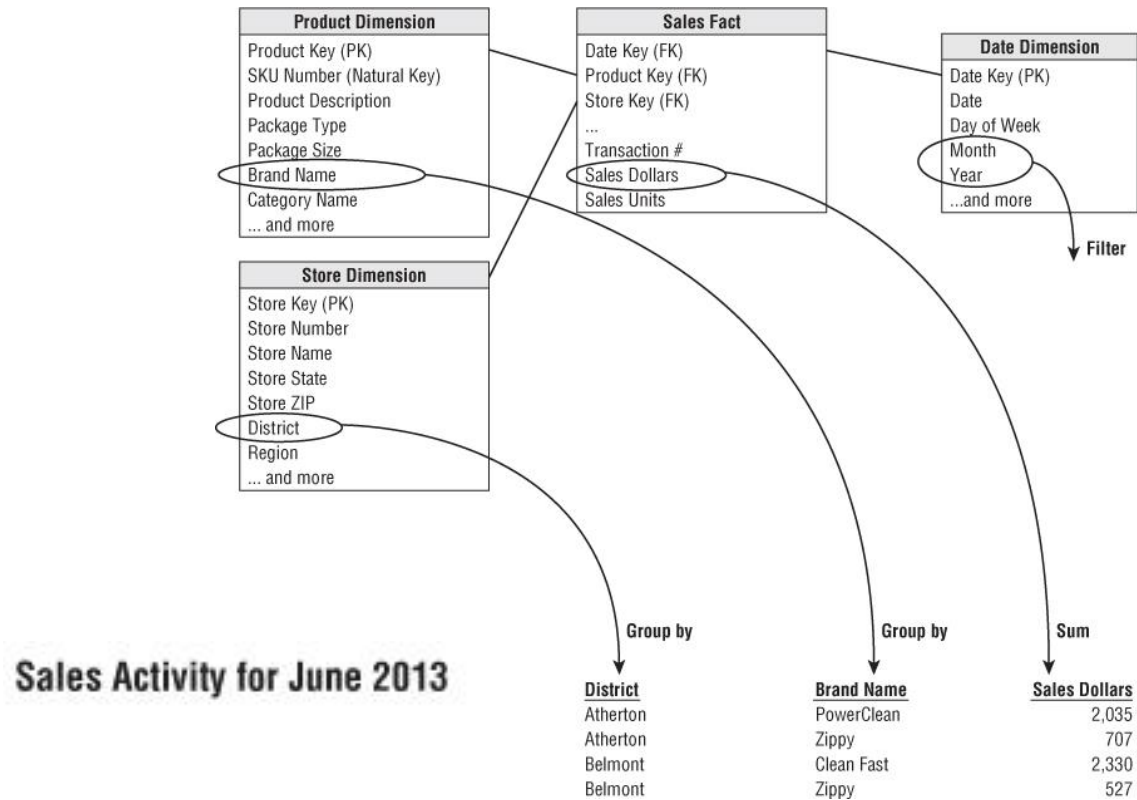


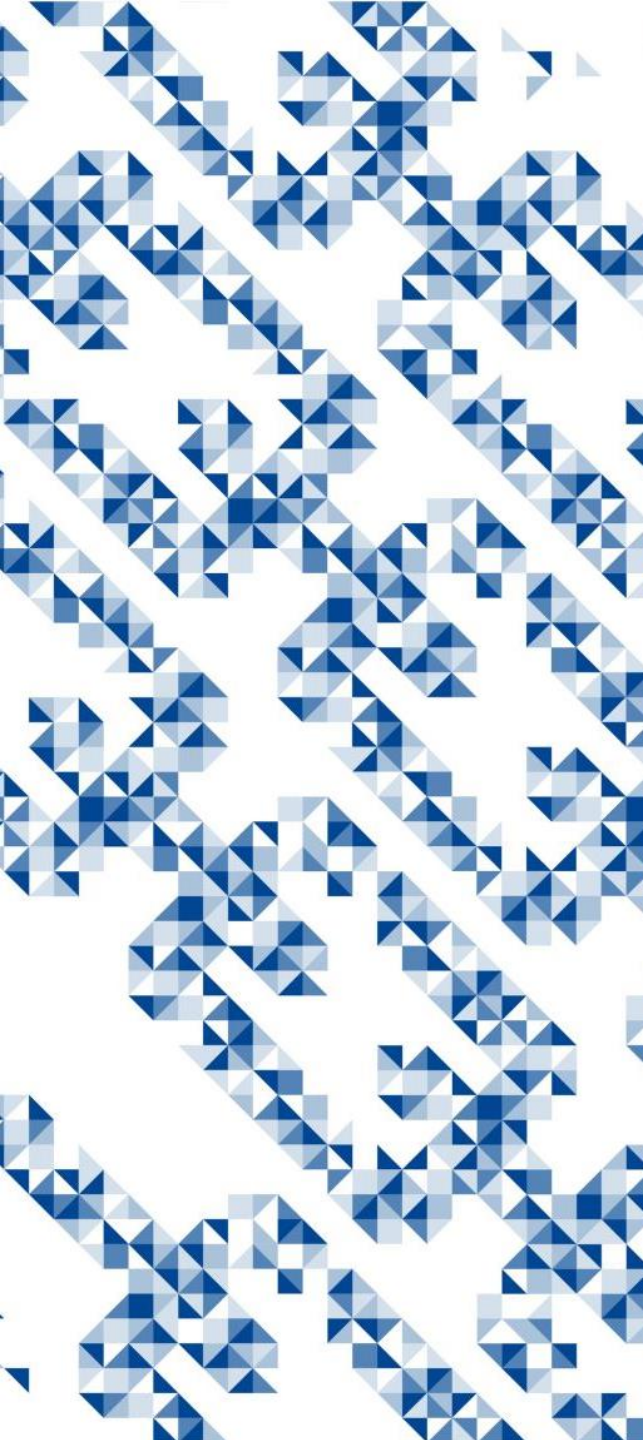
Star Schema

```
SELECT *  
  FROM SALES S, TIME T, PRODUCT P, LOCATION L  
 WHERE S.TIME = T.ID      AND  
        S.PRODUCT = P.ID  AND  
        S.LOCATION = L.ID AND  
        T.YEAR = 2005     AND  
        P.CLASS = 'AUDIO' AND  
        L.LOCATION = 'SAN JOSE';
```



Star Schema





Dimensions = meaning

Dimensions and their values add meaning in several ways:

- They are used to filter queries or reports.

- They are used to control the scope of fact aggregation.

- They are used to order or sort information.

- They accompany facts to provide context for the reports.

- They are used to define hierarchies that allow you to define grouping, partial summation, and summary.

 - internal organization of dimensions within the parent-child relationship*



Fact tables = measurements

Fact tables store measurement data.

They have many rows, but usually few columns.

Fact tables for a large enterprise can easily hold billions of rows.

For many star schemes, the fact table will represent well over 90 percent of the total disk space.

A fact table has a composite key consisting of the keys of the basic schema dimension tables.

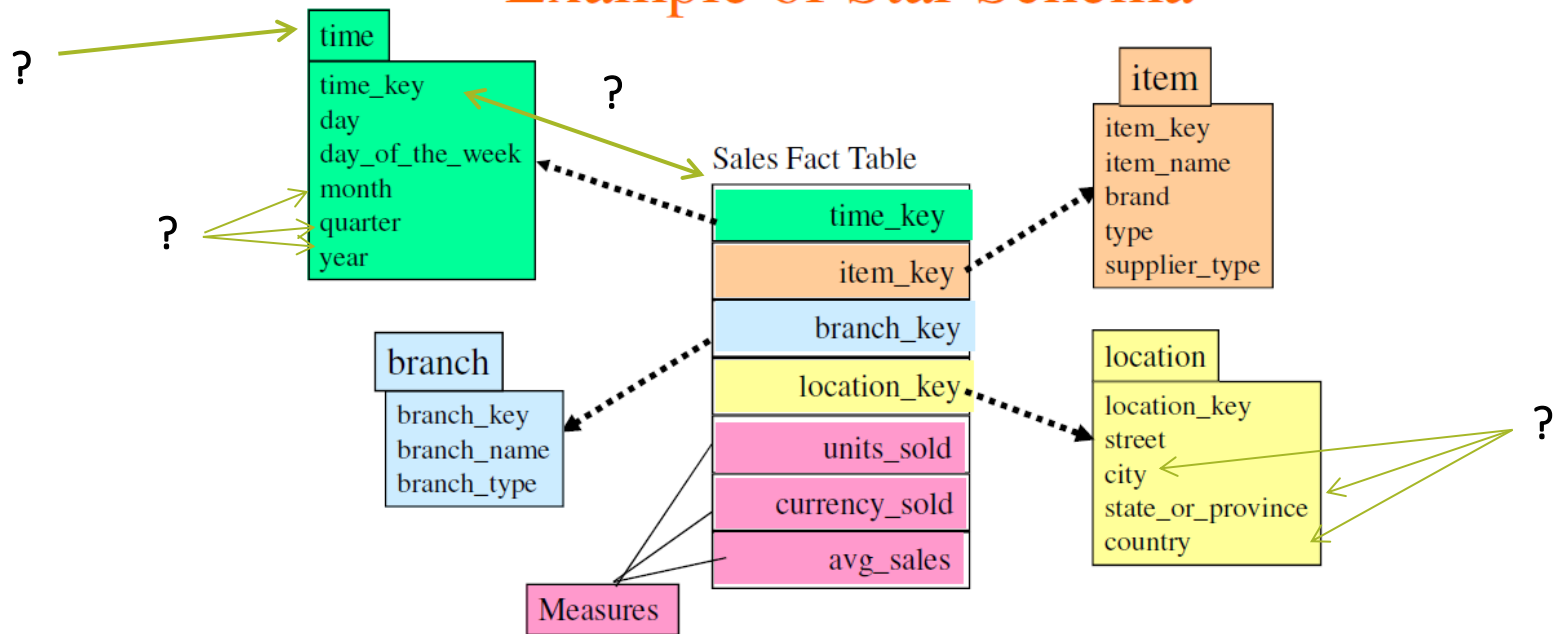
The fact table usually contains facts with the same level of aggregation

The fact table contains facts at the level of detail or facts that have been aggregated.

Fact tables that contain aggregated facts are often called summary tables.

Star Schema

Example of Star Schema



Star Schema

Advantages

Facts and dimensions are clearly depicted

dimension tables are relatively static, data is loaded (append mostly) into fact table(s)

Easy to comprehend (and write queries)

Less tables and simple structure

Disadvantages

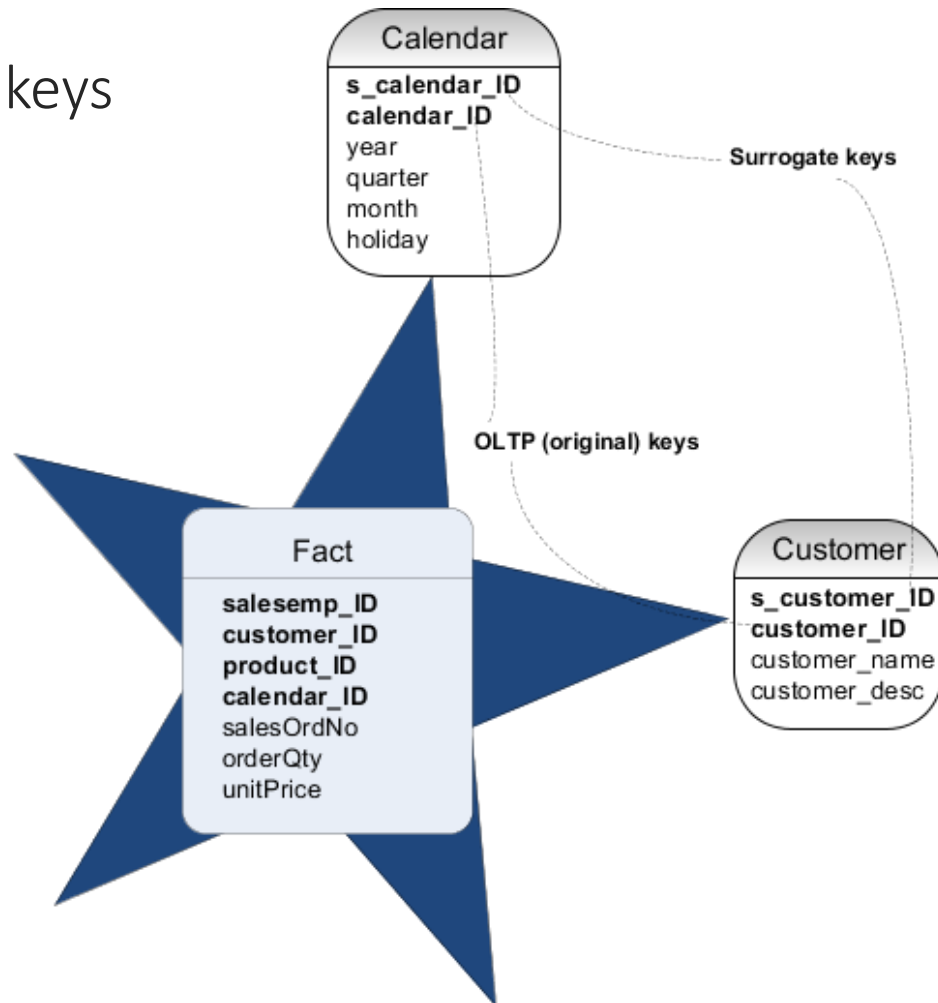
In some cases, high overhead of redundant data

dimension tables must carry a level indicator for every record and every query must use it

level is needed whenever aggregates are stored with detail facts

Star Schema

Surrogate keys

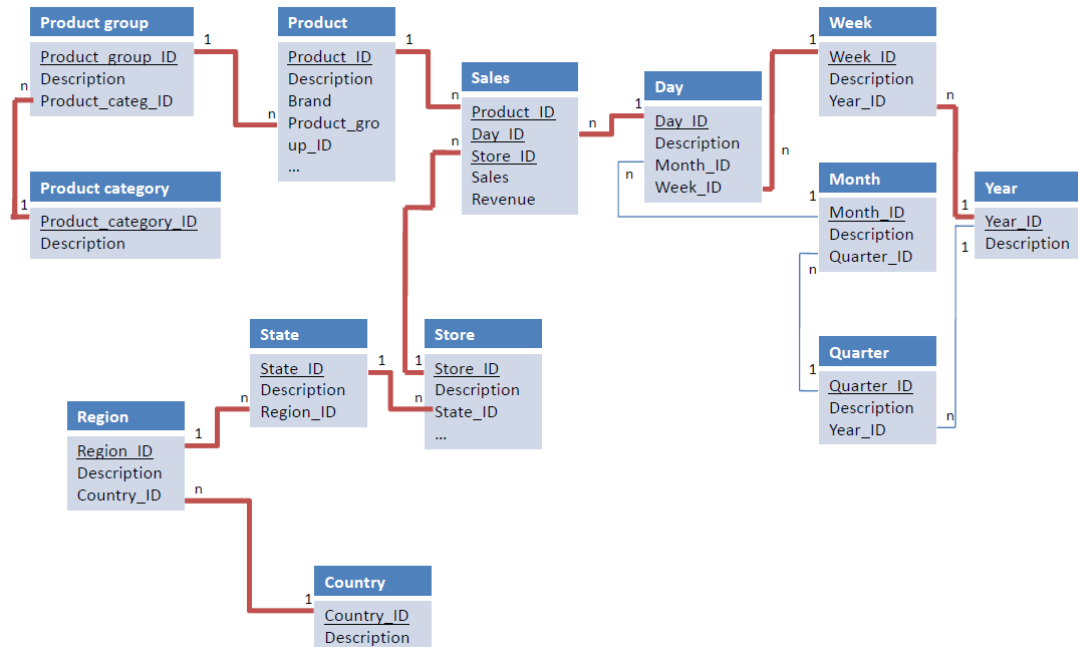


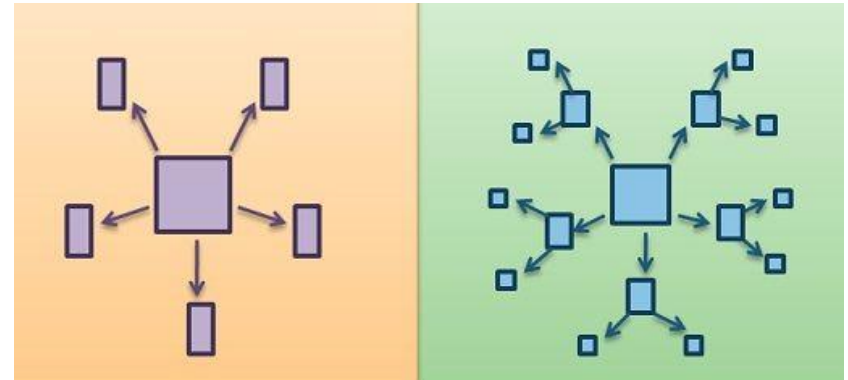
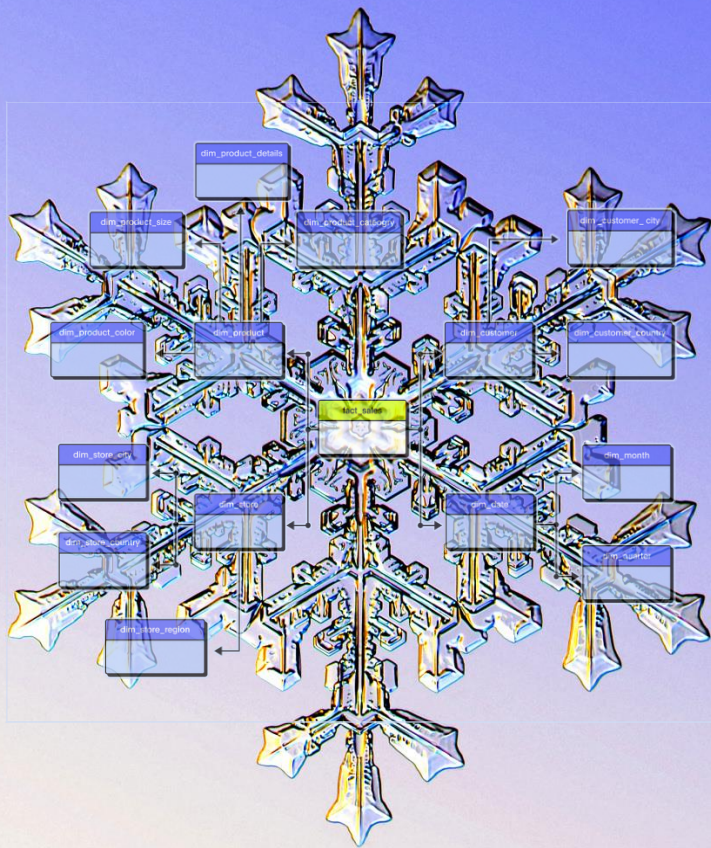
Snowflake schema

Snowflake schema:

The facts/measures are in the center

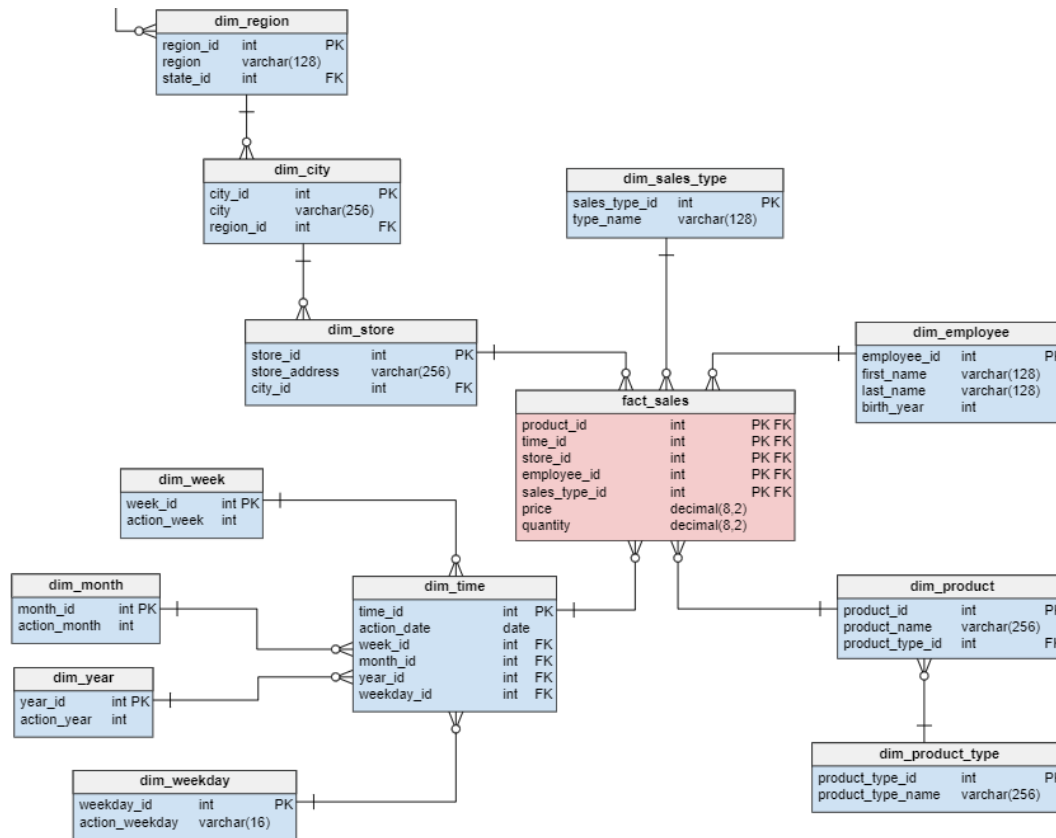
The dimensions spread out in each direction and branch out with their granularity





Snowflake schema

Snowflake schema



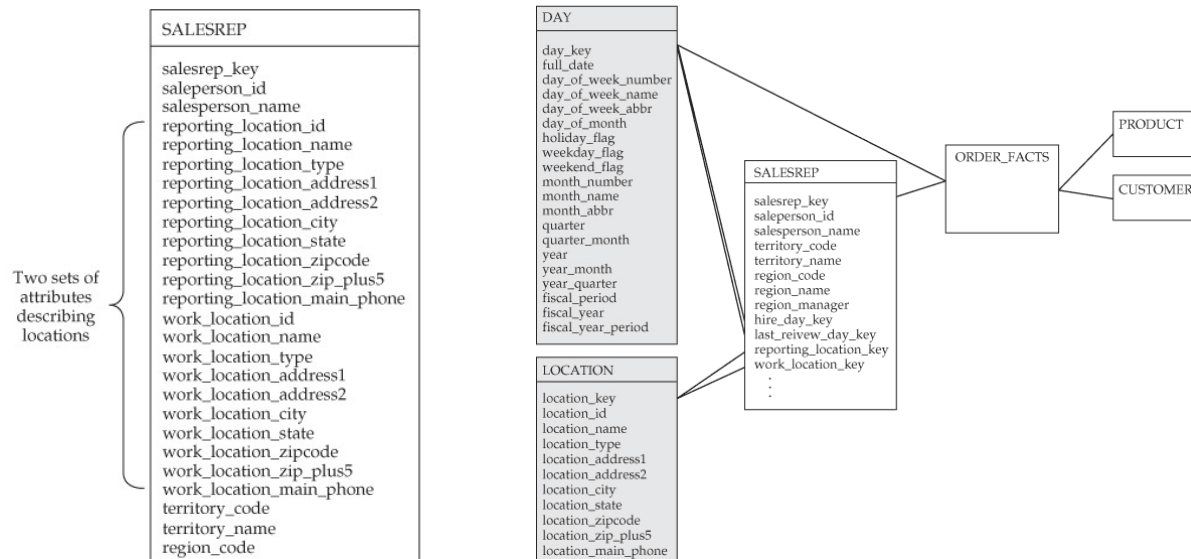
Snowflake schema

Snowflake schema:

One of the best reasons to design a snowflake schema is that your database or business intelligence tool requires it.

Repeating Groups

A group of attributes appears more than once in a dimension table



Snowflake schema

Snowflake schema:

One of the best reasons to design a snowflake schema is that your database or business intelligence tool requires it.

Missing attribute's values

A group of attributes is missing for a significant portion of data

Example

Medical clinic has two types of patients – regular and sporadic

Sporadic customers – only PESEL is known, other details are missing

If two types are nearly equal then we need to store a lot of missing values

In the denormalized Patient dimension

Snowflake schema

Snowflake schema:

One of the best reasons to design a snowflake schema is that your database or business intelligence tool requires it.

Multiple hierarchies

Many attributes

It's easier to represent all relations explicitly – each hierarchy in a separate table

Multiple levels

Each level in a separate table

Snowflake schema

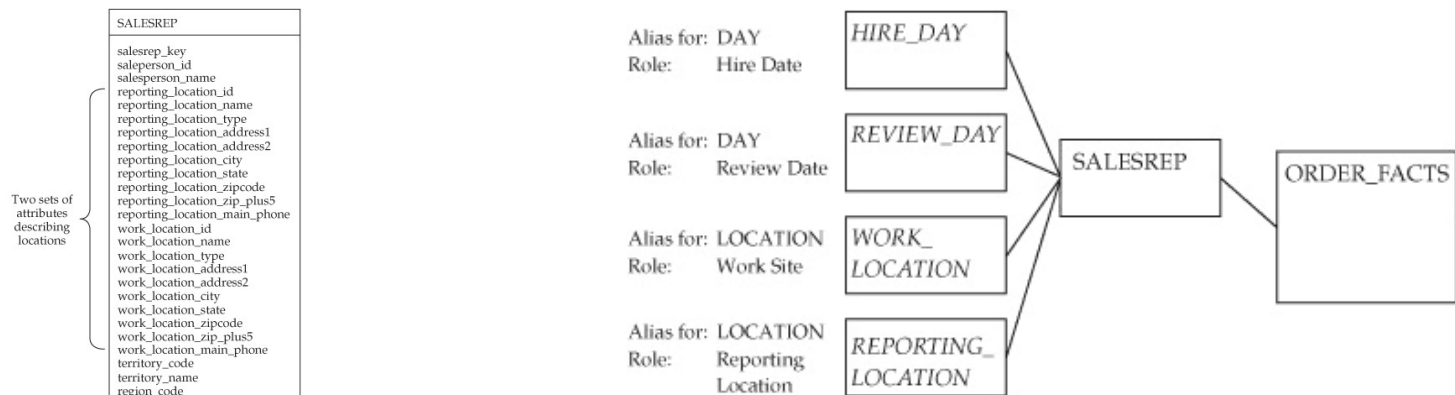
Snowflake schema:

Reasons to design a snowflake schema

your database or business intelligence tool requires it

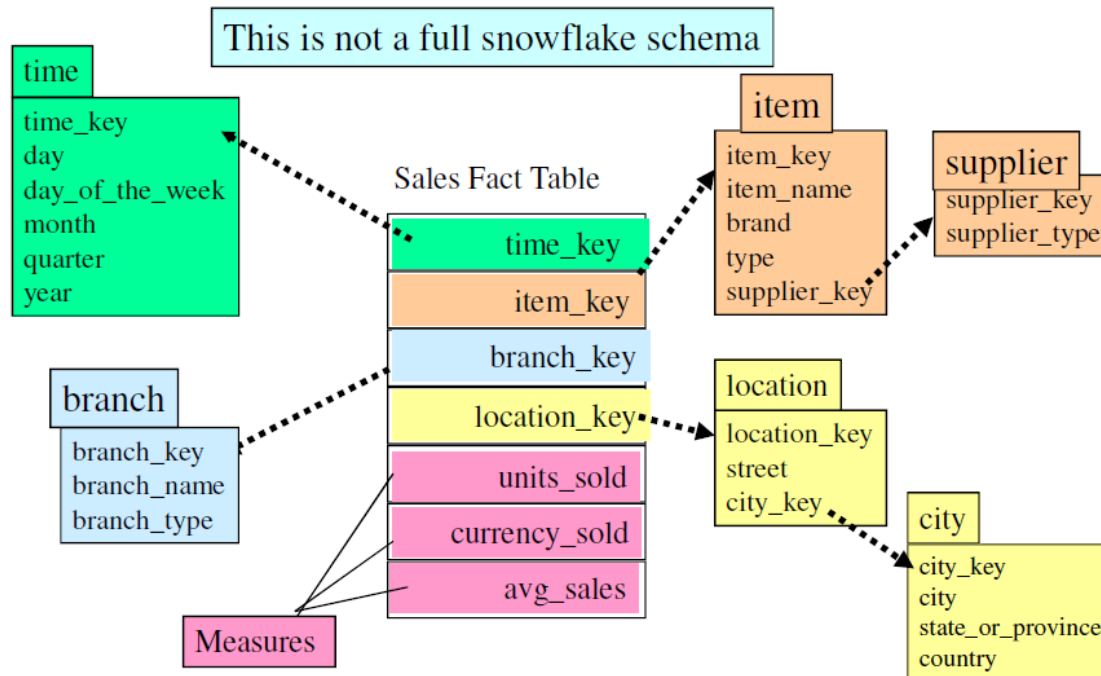
a group of attributes appears more than once in a dimension table

repeating attributes are placed in a new table, linked to the dimension via a primary key / foreign key relationship – called an outrigger



Snowflake schema

Example of Snowflake Schema



Snowflake schema

Snowflake schema:

A refinement of star schema where some dimensional hierarchy is normalized into a set of smaller dimension tables, forming a shape similar to snowflake

Advantages

With a snowflake schema the size of the dimension tables will be reduced and some queries might run faster

Easier to handle changes

Easier to handle ETL process

Disadvantages

Fact tables are responsible for 90% of the storage requirements

normalizing the dimensions usually leads to insignificant improvements

Normalization of the dimension tables can reduce the performance of the DW because it leads to a large number of tables

connecting dimensions with coarse granularity requires multiple joins

More complex

DimProduct	
ProductKey	
ProductAlternateKey	
ProductSubcategoryKey	
WeightUnitMeasureCode	
SizeUnitMeasureCode	
EnglishProductName	
SpanishProductName	
FrenchProductName	
StandardCost	
FinishedGoodsFlag	
Color	
SafetyStockLevel	
ReorderPoint	
ListPrice	
Size	
SizeRange	
Weight	
DaysToManufacture	
ProductLine	
DealerPrice	
Class	
Style	
ModelName	
LargePhoto	
EnglishDescription	
FrenchDescription	
ChineseDescription	
ArabicDescription	
HebrewDescription	
ThaiDescription	
GermanDescription	
JapaneseDescription	
TurkishDescription	
StartDate	
EndDate	
Status	

DimDate	
DateKey	
FullDateAlternateKey	
DayNumberOfWeek	
EnglishDayNameOfWeek	
SpanishDayNameOfWeek	
FrenchDayNameOfWeek	
DayNumberOfMonth	
DayNumberOfYear	
WeekNumberOfYear	
EnglishMonthName	
SpanishMonthName	
FrenchMonthName	
MonthNumberOfYear	
CalendarQuarter	
CalendarYear	
CalendarSemester	
FiscalQuarter	
FiscalYear	
FiscalSemester	

DimPromotion	
PromotionKey	
PromotionAlternateKey	
EnglishPromotionName	
SpanishPromotionName	
FrenchPromotionName	
DiscountPct	
EnglishPromotionType	
SpanishPromotionType	
FrenchPromotionType	
EnglishPromotionCategory	
SpanishPromotionCategory	
FrenchPromotionCategory	
StartDate	
EndDate	
MinQty	
MaxQty	

DimProductSubcategory	
ProductSubcategoryKey	
ProductSubcategoryAlternateKey	
EnglishProductSubcategoryName	
SpanishProductSubcategoryName	
FrenchProductSubcategoryName	
ProductCategoryKey	

DimProductCategory	
ProductCategoryKey	
ProductCategoryAlternateKey	
EnglishProductCategoryName	
SpanishProductCategoryName	
FrenchProductCategoryName	

FactInternetSales	
ProductKey	
OrderDateKey	
DueDateKey	
ShipDateKey	
CustomerKey	
PromotionKey	
CurrencyKey	
SalesTerritoryKey	
SalesOrderNumber	
SalesOrderLineNumber	
RevisionNumber	
OrderQuantity	
UnitPrice	
ExtendedAmount	
UnitPriceDiscountPct	
DiscountAmount	
ProductStandardCost	
TotalProductCost	
SalesAmount	
TaxAmt	
Freight	
CarrierTrackingNumber	
CustomerPONumber	
OrderDate	
DueDate	
ShipDate	

DimCustomer	
CustomerKey	
GeographyKey	
CustomerAlternateKey	
Title	
FirstName	
MiddleName	
LastName	
NameStyle	
BirthDate	
MaritalStatus	
Suffix	
Gender	
EmailAddress	
YearlyIncome	
TotalChildren	
NumberChildrenAtHome	
EnglishEducation	
SpanishEducation	
FrenchEducation	
EnglishOccupation	
SpanishOccupation	
FrenchOccupation	
HouseOwnerFlag	
NumberCarsOwned	
AddressLine1	
AddressLine2	
Phone	
DateFirstPurchase	
CommuteDistance	

DimSalesTerritory	
SalesTerritoryKey	
SalesTerritoryAlternateKey	
SalesTerritoryRegion	
SalesTerritoryCountry	
SalesTerritoryGroup	
SalesTerritoryImage	

DimGeography	
GeographyKey	
City	
StateProvinceCode	
StateProvinceName	
CountryRegionCode	
EnglishCountryRegionName	
SpanishCountryRegionName	
FrenchCountryRegionName	
PostalCode	
SalesTerritoryKey	
IpAddressLocator	

DimCurrency	
CurrencyKey	
CurrencyAlternateKey	
CurrencyName	

FactInternetSalesReason	
SalesOrderNumber	
SalesOrderLineNumber	
SalesReasonKey	



Snowflake Schema

Logical level design

- Some sources claim that logical design only uses star schema
- Whereas normalization of dimensions is a technique of transforming a logical model to physical model

In practice

- We start with a Star Schema
- Create the “snowflakes” only with queries.
 - *This eliminates the need to create separate extracts for each table*

Stars



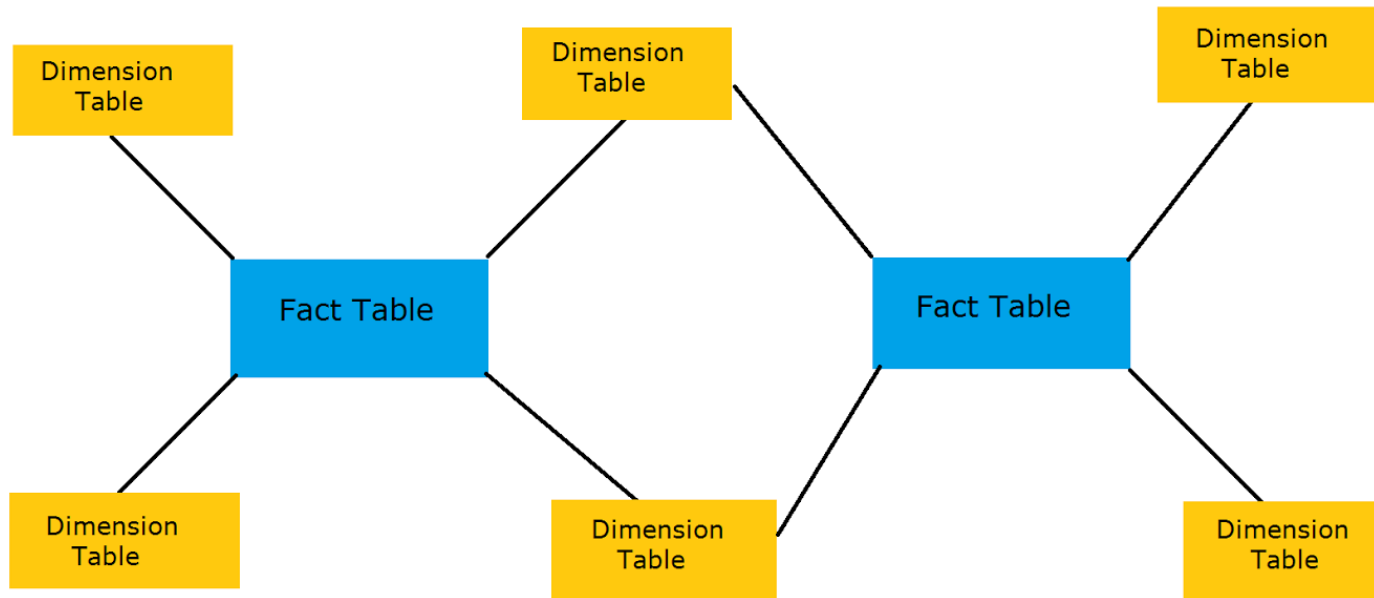
Each new star

- allows to analyse the business process measured
- allows the process to be studied in different context/perspectives

The stars that represent different processes can connect to one another through a set of common dimensions

- allows the process to be studied in conjunction with other processes
 - *cross-process analysis*
- nearly any two fact tables that link to the same dimension can theoretically be compared using the drill-across technique

Fact Constellation Schema



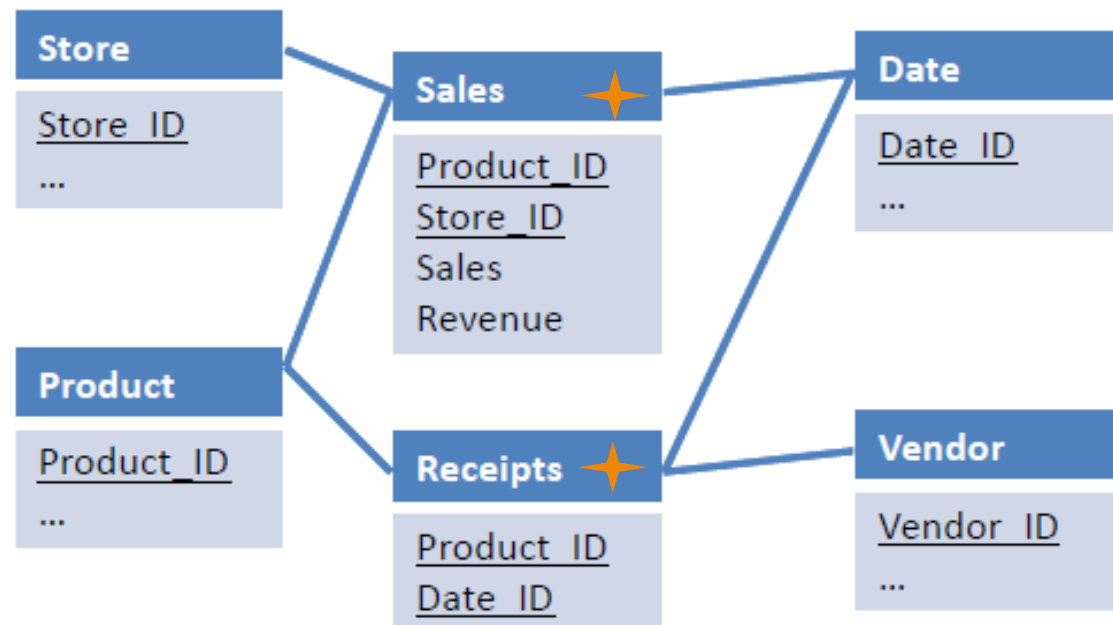
Fact Constellation Schema

Fact constellations:

multiple fact tables share dimension tables

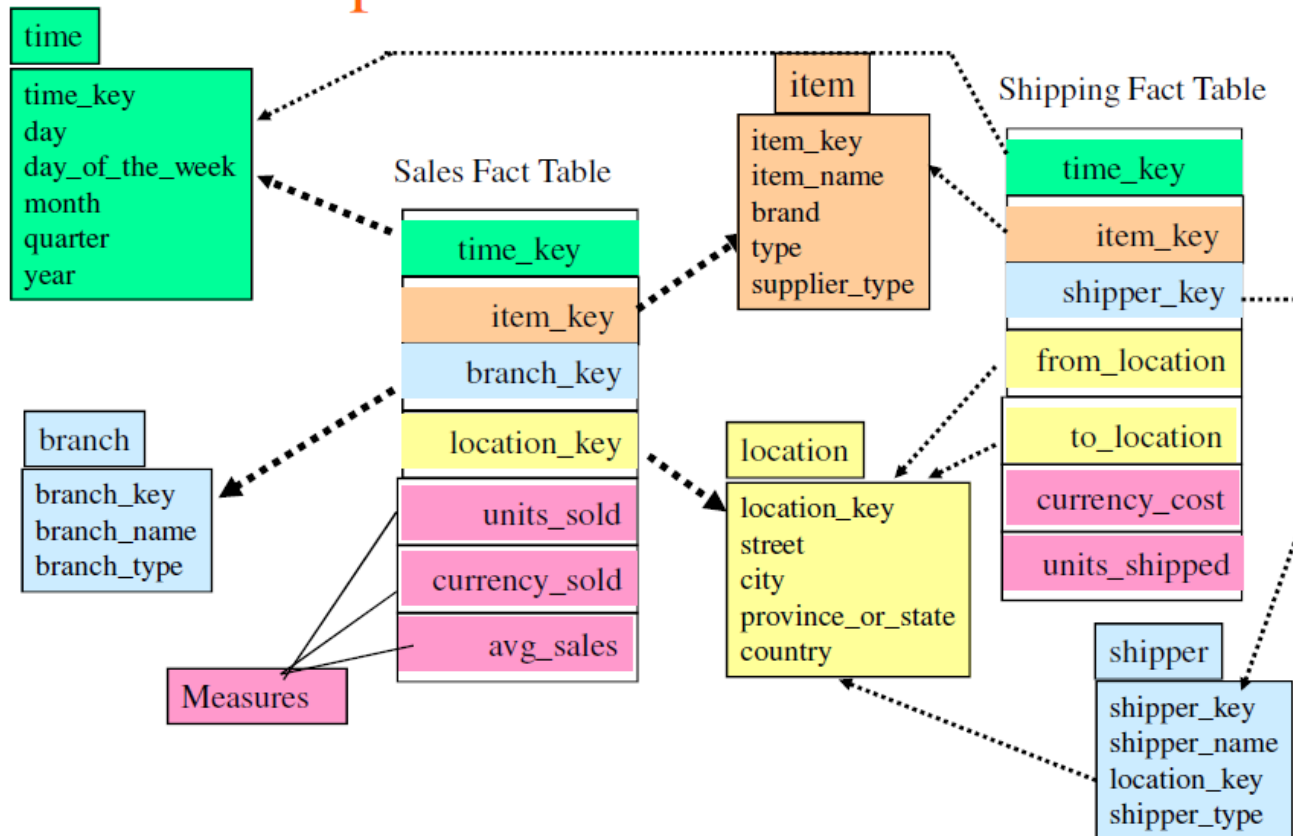
viewed as a collection of stars, therefore often called:

a galaxy schema or a fact constellation schema



Fact Constellation Schema

Example of Fact Constellation



Fact Constellation Schema

Drill-across:

Sales:

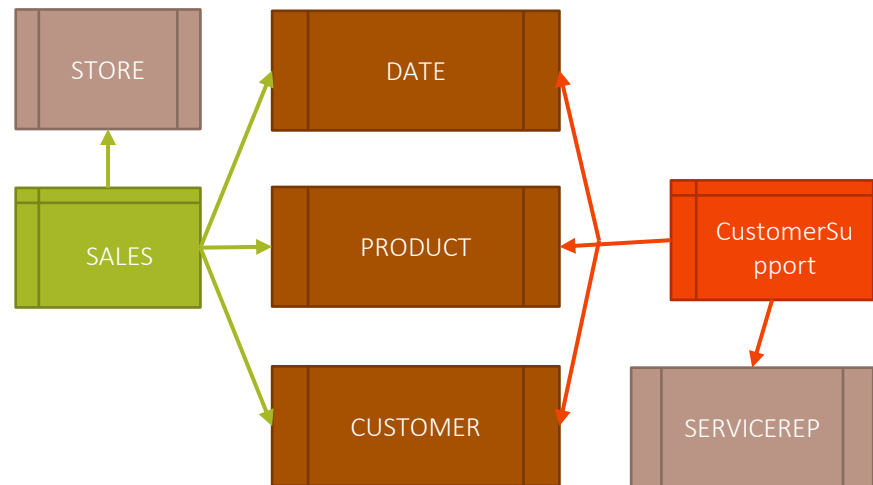
Date, Customer, Product, Store

CustomerSupport:

Date, Customer, ServiceRep

Query

How does the frequency of customer service contacts in region Y affect sales of product X?



Fact Constellation Schema

Query

How does the frequency of customer service contacts in region Y affect sales of product X?

Implementation:

Step 1: Query to CustomerSupport

*Group by CustomerID ; Filter per region = Y;
Calculate COUNT*

Query scheme

CustomerID, SupportCallCount

Step 2: Query Sales

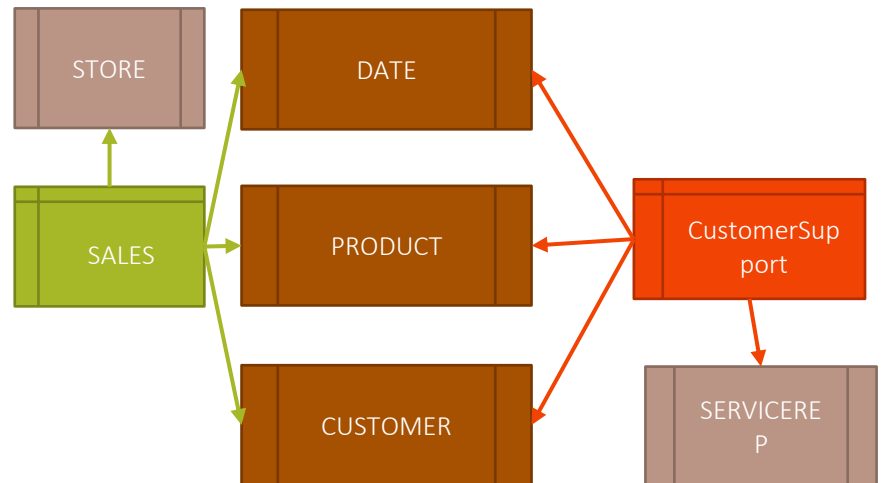
*Group by CustomerID ; Filter per region = Y,
product name = Product X ; Obliczenia
SUM(TotalSalesAmt)*

Query scheme

CustomerID, TotalSalesAmt

Step 3: Combine the results

*Connection based on CustomerID ; Group by
SupportCallCount ; COUNT and
AVG(TotalSalesAmt) calculations*



Fact Constellation Schema

Conformed dimensions

- Used by many fact tables

- They must retain their meanings in the context of various facts.

- They must be identical or one must be part of the other

Three types of conformed dimensions:

- The same

 - Dimension tables that are identical*

- Subset of rows

 - A dimension table has a subset of rows*

 - can increase performance when multiple rows are not relevant to a specific process

- Subset of attributes

 - A dimension table has a subset of attributes*

 - allows you to roll up dimensions with different grains

OLAP Server

A multidimensional database is a form of database in which data is stored in a multidimensional model

- cell contains values

 - the cell represents a snapshot, a particular measurement of a business process*

- coordinates of each cell

 - are defined by the number of hierarchical dimensions*

The term multidimensional database is often confused with the term OLAP (analytical processing):

- MDB is a database

- OLAP is an activity/process used to analyze a database

The confusion is caused by the word OLAP cube

- OLAP cube has the same meaning as MDB

 - means a multidimensional database*

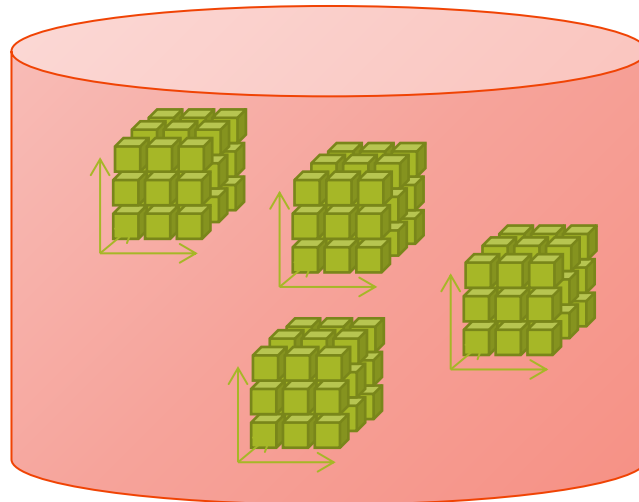
OLAP Server

Multidimensional DB (MDB)

Optimized for DW and OLAP applications

created using input from the staging area

designed for efficient and convenient storage and retrieval of large volumes of data



OLAP Server

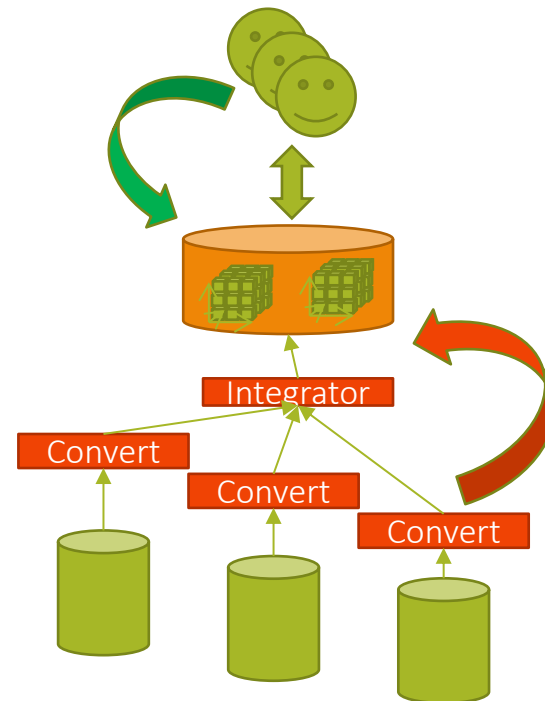
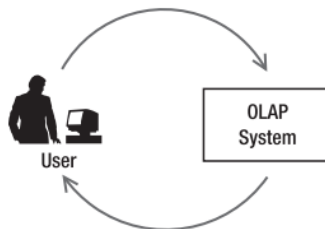
What do we use to manage multidimensional databases?

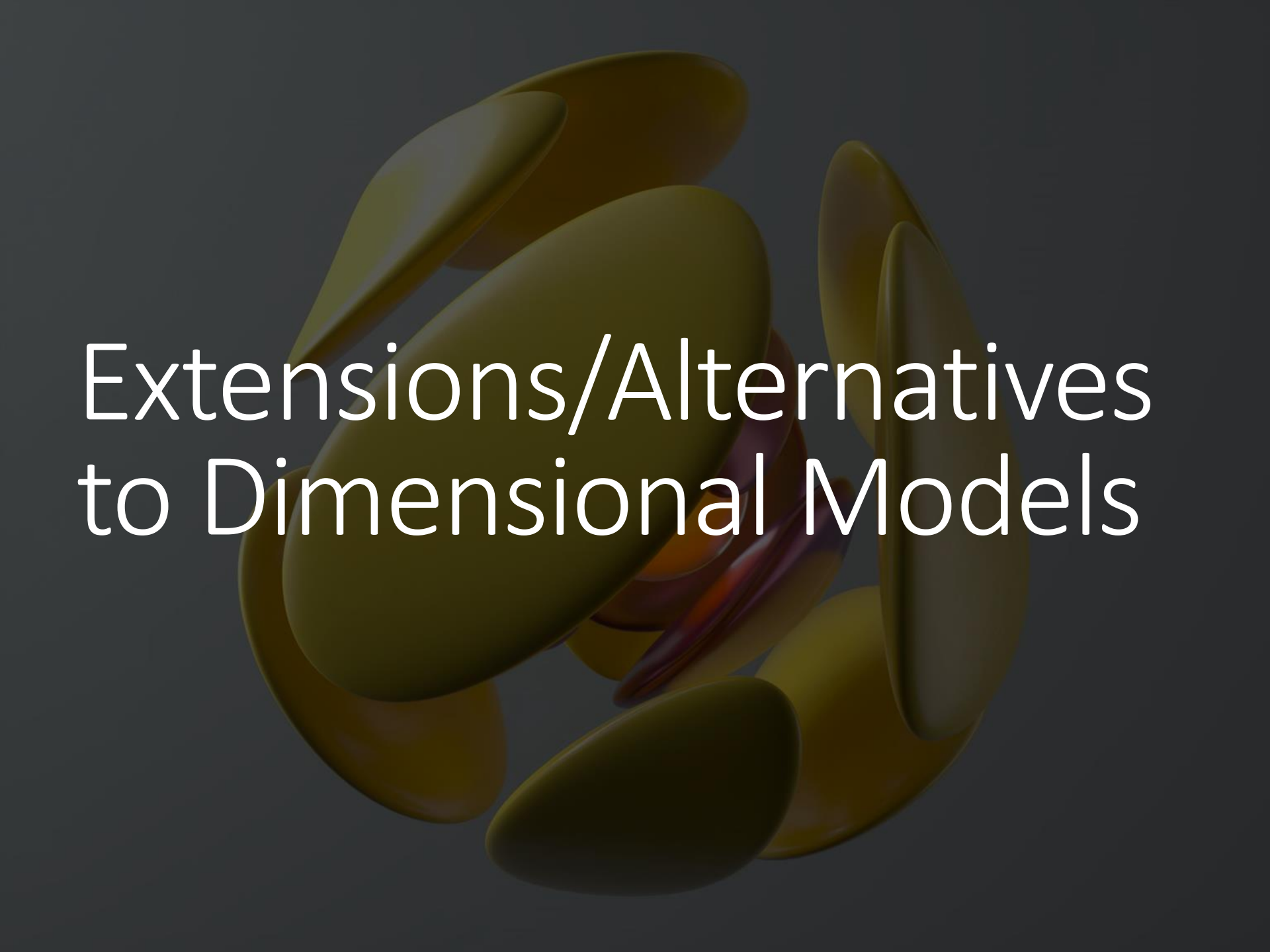
The system that manages and operates multidimensional databases is called a multidimensional database management system (MDBMS)

known as OLAP server or OLAP engine

The standard interface that connects to MDBMS is XML for analysis

known as XMLA



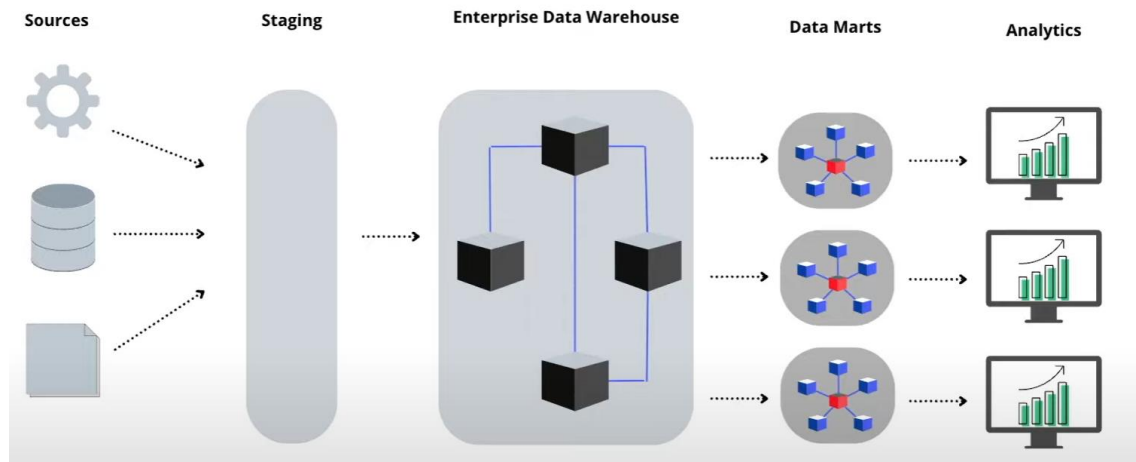


Extensions/Alternatives to Dimensional Models

Normalized

Normalized Modeling

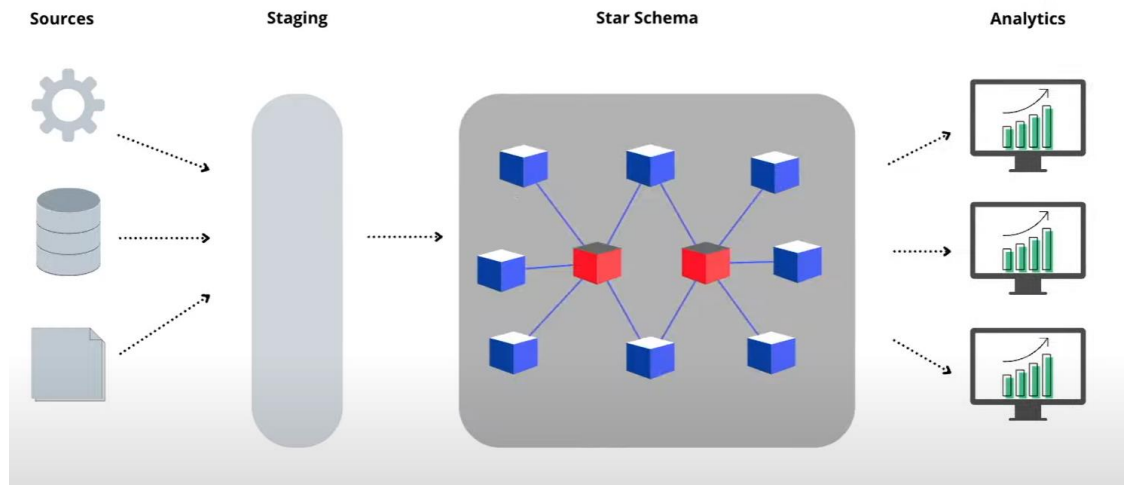
Bill Inmon

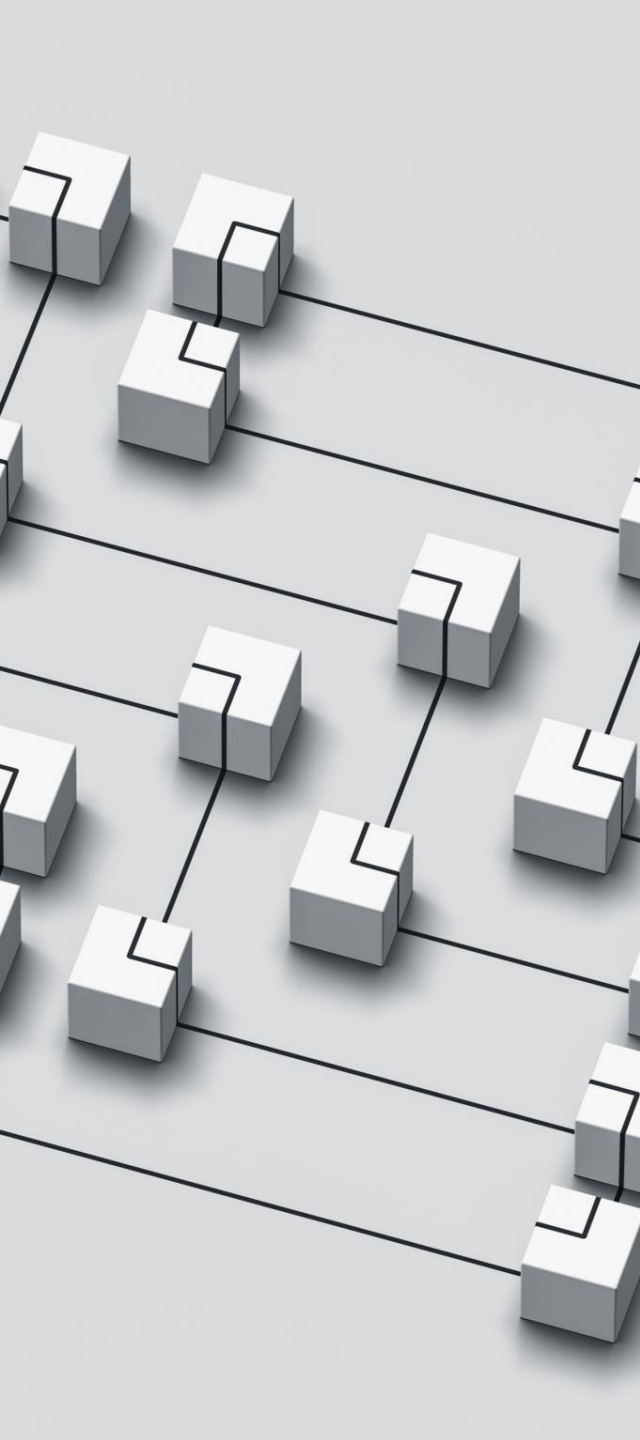


Dimensional

Denormalized Modeling

Ralph Kimball





Dimensional models

Dimensional models are resilient when data relationships change.

All the following changes can be implemented without altering any existing BI query or application, and without any change in query results:

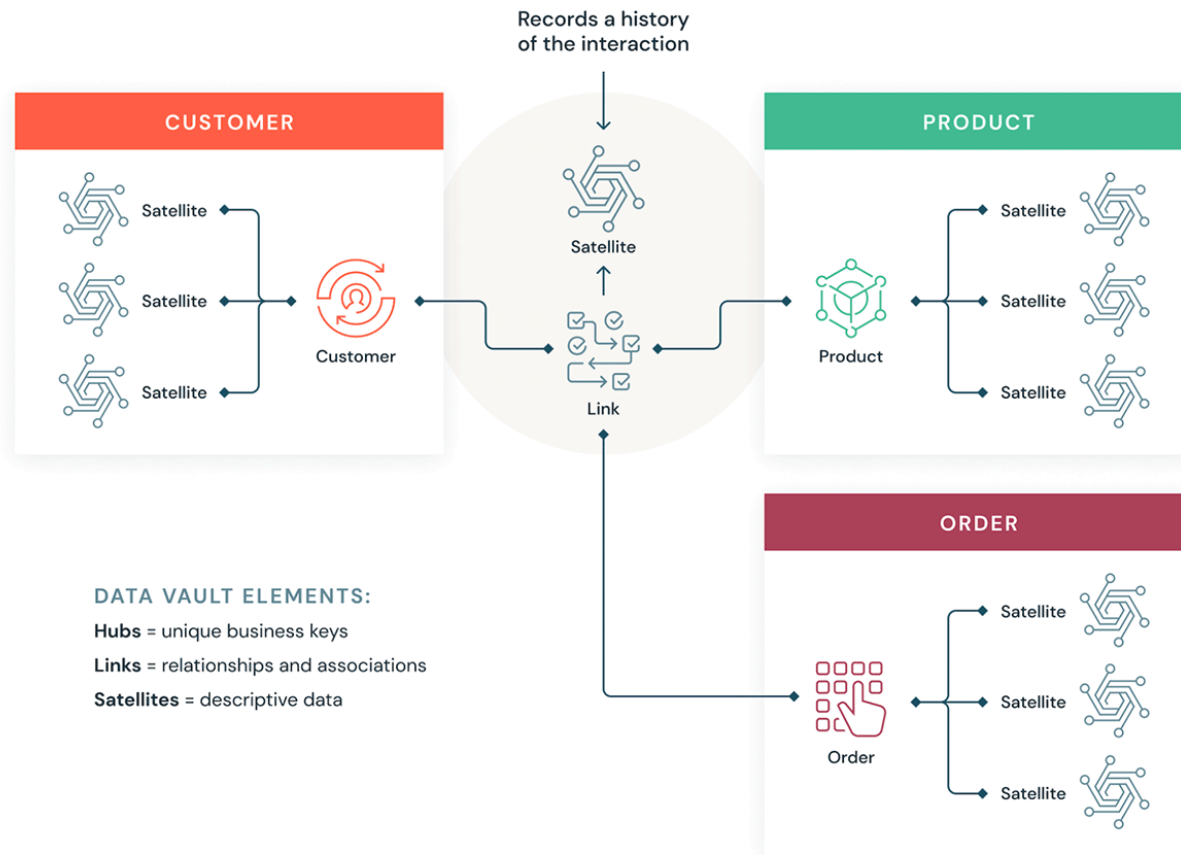
- Facts consistent with the grain of an existing fact table can be added by creating new columns.

- Dimensions can be added to an existing fact table by creating new foreign key columns

 - presuming they don't alter the fact table's grain.*

- Attributes can be added to an existing dimension table by creating new columns.

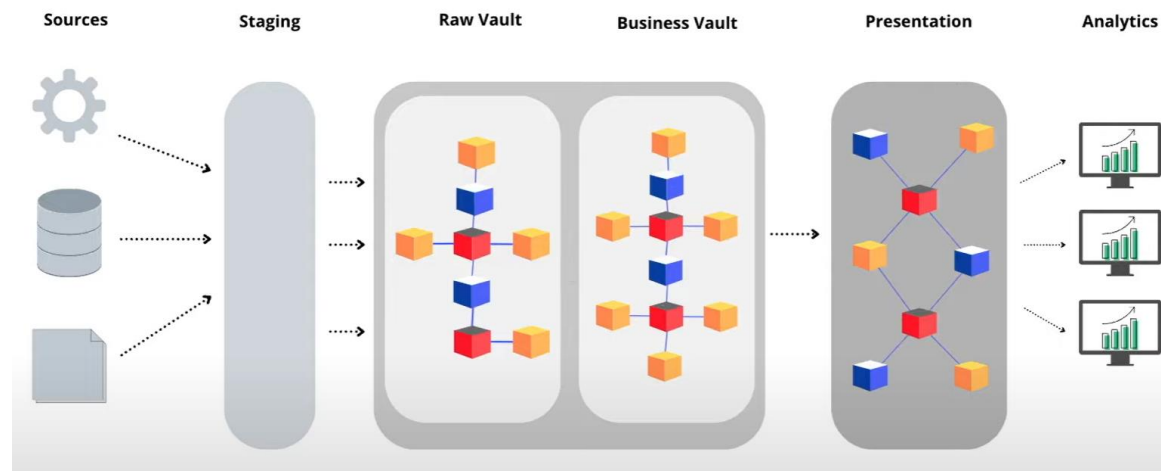
Data Vault



Data Vault

Data Vault 2.0

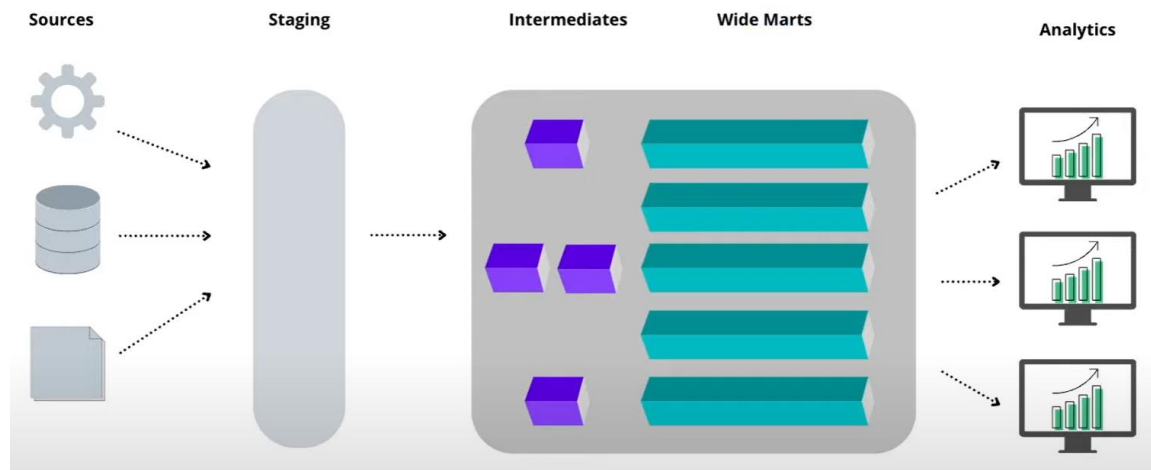
Hubs, Links & Satellites



OBT

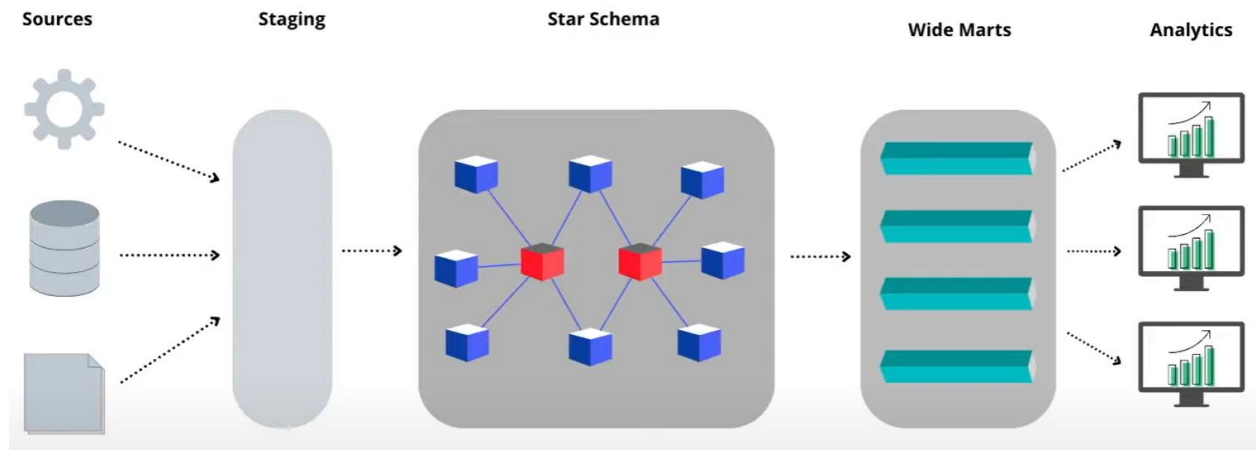
One-Big-Table (OBT)

Flatten Straight to Wide Tables



Hybrid

Hybrid Approach



Languages & Data Types

SQL

Python

Jinja

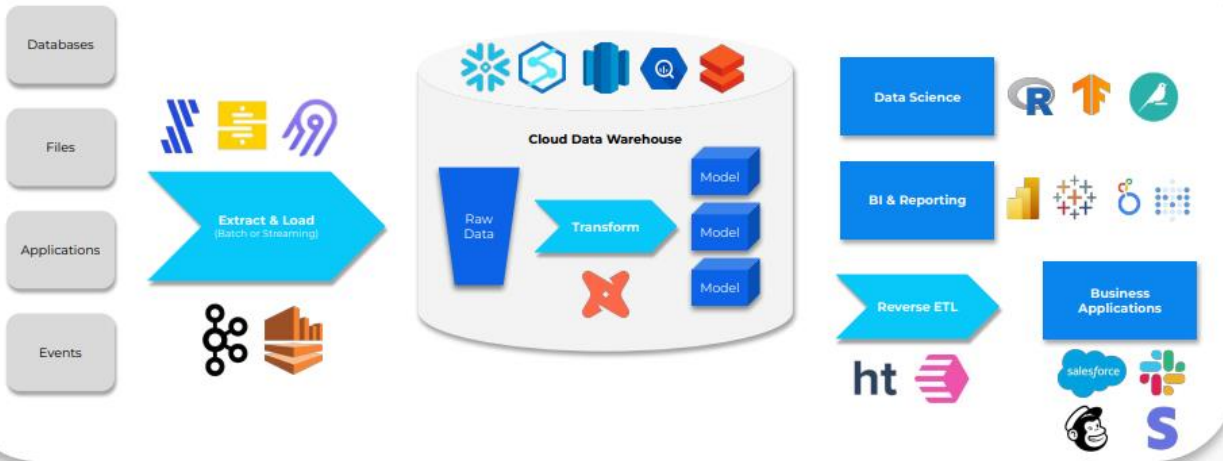
JSON/XML

YAML

DAX

Bash / PowerShell

Source Data



Analytics Data

Cloud Providers



Data Quality



Version Control & CI/CD



Infrastructure as Code (IaC)



Containerization



Container Orchestration



Scheduling & Task Orchestration



Surrounding Components

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Inmon W.,
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Delen, David King, Janine E. Aronson
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